# EXERCISE (O-1)

## SINGLE CORRECT TYPE QUESTIONS

### *Lattice Enthalpy; Condition for Formation of Ionic Bond*

**1.** Lattice enthalpy is the change in energy that occurs when ..... of an ionic solid is separated into isolated ions in the gas phase.
(A) One mole (B) One gram (C) Two mole (D) All of these
**CCB095**

**2.** Which of the following compounds will show the highest lattice energy?
(A) KF (B) NaF (C) CsF (D) RbF
**CCB097**

**3.** The lattice energy of the lithium compounds is in the following order :
(A) LiF > LiCl > LiBr > LiI (B) LiCl > LiF > LiBr > LiI
(C) LiBr > LiCl > LiF > LiI (D) LiI > LiBr > LiCl > LiF
**CCB098**

**4.** The order of increasing lattice energy of the following compounds, is :
(A) NaCl < CaO < NaBr < BaO (B) NaBr < NaCl < BaO < CaO
(C) NaCl > NaBr < BaO < CaO (D) NaBr < NaCl < CaO < BaO
**CCB103**

### *Covalent Bond; Types of Electron Pair; Bond Pair and Lone Pair*

**5.** Covalent bond is formed by :
(A) Combination of ions (B) Combination of half filled orbitals
(C) Exchange of electrons by atom (D) None of above
**CCB116**

**6.** The compound which contains ionic as well as covalent bonds is
(A) $C_2H_4Cl_2$ (B) $CH_3I$ (C) KCN (D) $H_2O_2$
**CCB121**

### *Coordinate Bond and Types of Coordinate Bond*

**7.** The bonds present in $N_2O_5$ are :
(A) Only ionic (B) Covalent & coordinate
(C) Only covalent (D) Covalent & ionic
**CCB006**

**8.** Which of the following is an ionic compound?
(A) $CH_4$ (B) $H_2O$ (C) $CaCl_2$ (D) $NH_3$
**CCB113**

**9.** The pair of compounds which can form a co–ordinate bond is :
(A) $(C_2H_5)_3B$ and $(CH_3)_3$ N (B) HCl and HBr
(C) $BF_3$ and $NH_3$ (D) (A) & (C) both
**CCB124**

**10.** The types of bond present in $N_2O_4$ are
(A) only covalent (B) only ionic (C) ionic and covalent (D) covalent & coordinate
**CCB125**

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**11.** Which of the following molecule does not have coordinate bonds?

(A) $CH_3$–NC (B) CO (C) $O_3$ (D) $CO_3^{2\ominus}$

**CCB126**

### *Kossel -Lewis approach to Chemical Bonding; Lewis Dot Structure*

**12.** The total number of lone pairs in a chlorine molecule is :

(A) Six (B) Three (C) Four (D) Seven

**CCB118**

**13.** The bond that exists between $NH_3$ and $BF_3$ is called

(A) Electrovalent (B) Covalent

(C) Coordinate (D) Hydrogen

**CCB130**

**14.** Which has a coordinate bond (draw LDS)

(A) $SO_3^{2\ominus}$ (B) $CH_4$ (C) $CO_2$ (D) $NH_3$

**CCB131**

**15.** The number of dative bonds in nitric acid molecule is :

(A) 0 (B) 1 (C) 2 (D) 4

**CCB133**

**16.** Indicate the nature of bonding in $CCl_4$ and $CaH_2$

(A) Covalent in $CaH_2$ and electrovalent in $CaH_2$

(B) Electrovalent in both $CCl_4$ and $CaH_2$

(C) Covalent in both $CCl_4$ and $CaH_2$

(D) Electrovalent in $CCl_4$ and covalent in $CaH_2$

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**CCB134**

**17.** Which species does not exists :

(A) $AlF_6^{3\ominus}$ (B) $BF_4^{\ominus}$ (C) $BeF_4^{2\ominus}$ (D) $CCl_6^{2\ominus}$

**CCB137**

**18.** According to LDS, dative bond is present in :

(A) $SO_3$ (B) $NH_3$ (C) $K_2CO_3$ (D) $BF_3$

**CCB138**

### *Formal Change*

**19.** The formal charges on the three O-atoms in $O_3$ molecule are:

(A) 0, 0, 0 (B) 0, 0, –1 (C) 0, 0, +1 (D) 0, +1, –1

**CCB217**

### *Limitations of Lewis Dot Structure (Hypervalent ,Hypovalent , Odd Electron Species)*

**20.** The octet rule is not valid for the molecule :

(A) $CO_2$ (B) $NH_3$ (C) $SF_2$ (D) $AlCl_3$

**CCB142**

**21.** Which of the following pair has electron deficient compounds ?

(A) $BCl_3$, $AlCl_3$ (B) $C_2H_6$, $PCl_6^{\ominus}$

(C) $SF_2$, $Cl_2O$ (D) $BF_4^{\ominus}$, ICl

**CCB143**



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**22.** Which is not an exception to octet rule ?
(A) $BF_3$ (B) $SnCl_4$ (C) $BeI_2$ (D) $ClO_2$

**CCB144**

**23.** Which of the following is an example of hypovalent species?
(A) $BeCl_2$ (B) $PCl_5$ (C) $CO_2$ (D) $SF_6$

**CCB146**

**24.** Which of the following species are hypervalent?
(1) $PCl_5$, (2) $BF_3$, (3) $XeF_2$, (4) $CO_3^{2\ominus}$
(A) 1, 2, 3 (B) 1, 3 (C) 3, 4 (D) 1, 2

**CCB147**

### *Introduction to Valence Bond Theory; Covalency*

**25.** In allene structure ($H_2C=C=CH_2$) three carbon atoms are joined by :
(A) Three sigma bonds and three pi bonds
(B) Two sigma bonds and one pi bond
(C) Two sigma bonds and two pi bonds
(D) Three pi bonds only

**CCB119**

**26.** Number of σ and π bonds present in $CH_3 - CH = CH - C \equiv CH$ are :
(A) 10σ, 3π (B) 10σ, 2π (C) 9σ, 2π (D) 8σ, 3π

**CCB120**

**27.** Variable covalency is exhibited by :
(A) P and S (B) N and O (C) N and P (D) F and Cl

**CCB115**

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**28.** In which of the excitation state of Chlorine, $ClF_3$ is formed?
(A) In ground state (B) In third excitation state
(C) In first excitation state (D) In second excitation state

**CCB158**

**29.** Which compound of xenon is not possible?
(A) $XeF_2$ (B) $XeF_4$ (C) $XeF_5$ (D) $XeF_6$

**CCB159**

**30.** Nitrogen does not form $NF_5$ because :
(A) Nitrogen is member of V group (B) It contains no empty d-orbital
(C) The bond energy of $N \equiv N$ is very high (D) Fluorine is highly electronegative

**CCB160**

**31.** Which of the following compound is formed in the second excitation state of sulphur atom ?
(A) $SF_4$ (B) $SF_6$ (C) $SF_2$ (D) None

**CCB161**

### *Types of Covalent Bond and Overlapping of Orbitals; Orbital Overlap along Different Axes*

**32.** A sigma bond is formed by the overlapping of :
(A) s-s orbital alone
(B) s and p orbitals alone
(C) s–s, s–p or p–p orbitals along internuclear axis
(D) p–p orbital along the sides

**CCB163**



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**33.** The ratio of number of σ bond to π bond in $N_2$ and CO molecules are :

(A) 2.0, 2.0 (B) 2, $\frac{1}{2}$ (C) $\frac{1}{2}, \frac{1}{2}$ (D) $\frac{1}{2}$, 2

**CCB166**

**34.** No. of σ and π bonds in $C_2(CN)_4$ are respectively :

(A) 9σ, 9π (B) 8σ, 7π (C) 1σ, 1π (D) 9σ, 8π

**CCB167**

**35.** If x – axis is the approaching axis between two atoms, then which of the set of orbitals can form π-bond between two atoms in general ?

(A) s + px (B) $p_y + p_y$ (C) $p_x + p_x$ (D) $p_y + p_z$

**CCB168**

**36.** Which of the set of orbitals can form 'δ' bond between two atoms?

(A) $d_{yz} + d_{yz}$ along y – axis. (B) $d_{yz} + d_{yz}$ along z – axis.

(C) $d_{yz} + d_{yz}$ along x – axis. (D) $d_{xz} + d_{xz}$ along x – axis.

**CCB169**

**37.** Select the **CORRECT** statement if internuclear axis is y-axis :-

(A) $d_{xy}$ and $d_{xy}$ orbital of two atoms form π-bond

(B) $p_z$ and $p_z$ orbital of two atoms form σ-bond

(C) $d_{x^2-y^2}$ and $d_{x^2-y^2}$ orbital of two atoms form π-bond

(D) $p_y$ and $d_{zx}$ orbital of two atoms form π-bond

**CCB177**

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**38.** The number of sigma and pi bonds in 1-butene-3-yne are

(A) 5 sigma and 5 pi (B) 7 sigma and 3 pi

(C) 8 sigma and 2 pi (D) 6 sigma and 4 pi

**CCB178**

## *Order of Relative Strength of Covalent Bonds*

**39.** Nitrogen forms $N_2$ but phosphorus do not forms $P_2$, but it exists as $P_4$. The reason for this is :

(A) Triple bond is present between phosphorus atoms

(B) pπ - pπ bonding is weak

(C) pπ - pπ bonding is strong

(D) Multiple bond is formed easily

**CCB171**

**40.** The correct order of decreasing bond energy is :

(A) O–O > S – S > Se – Se (B) C – C > Si – Si > Ge – Ge

(C) F – F > O – O > N – N (D) F – F > Cl – Cl > Br – Br

**CCB172**

**41.** The strength of bonds by 2s - 2s, 2p - 2p and 2p -2s overlapping has the order :

(A) 2s – 2s > 2p – 2p > 2s – 2p (B) 2s – 2s > 2p – 2s > 2p – 2p

(C) 2p – 2p > 2s – 2p > 2s – 2s (D) 2p – 2p > 2s – 2s > 2p – 2s

**CCB173**

**42.** Which of the following bond is strongest?

(A) 1s – 1s (B) 2p – 2p (C) 2s – 2p (D) 1s – 2p

**CCB174**



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**43.** Consider the following statements :
I. A sigma ($\sigma$) bond is formed when two s-orbitals overlap
II. A pi($\pi$) bond is formed when two p-orbitals axially overlap
III. A $\sigma$ - bond is weaker than $\pi$ - bond
Which of the above statements is/are correct ?
(A) I and II (B) II and III (C) I alone (D) II alone

**CCB175**

### *Need of Hybridisation; Introduction of Hybridisation*

**44.** Which of the following are arranged in increasing order of length of the hybrid orbitals ?
(A) $sp < sp^2 < sp^3$ (B) $sp^3 < sp^2 < sp$ (C) $sp^2 < sp^3 < sp$ (D) $sp^2 < sp < sp^3$

**CCB183**

**45.** Which of the following is incorrect about hybridization ?
(A) The concept of hybridization is not applied to isolated atoms.
(B) Hybridization is the mixing of at least two non-equivalent atomic orbitals.
(C) The number of hybrid orbitals generated is more than the number of pure atomic orbitals that participate in the hybridization process.
(D) Number of hybrid orbitals formed is same as that of number of orbitals participate in mixing.

**CCB186**

**46.** In the context of carbon, which of the following is arranged in the correct order of electronegativity :
(A) $sp > sp^2 > sp^3$ (B) $sp^3 > sp^2 > sp$ (C) $sp^2 > sp > sp^3$ (D) $sp^3 < sp < sp^2$

**CCB197**

**47.** Select the **CORRECT** set of statements :
As the s-character of a hybrid orbital decreases
(I) The bond angle decreases (II) The bond strength increases
(III) The bond length increases (IV) Size of orbital increases
(A) (I), (III) and (IV) (B) (II), (III) and (IV) (C) (I) and (II) (D) All are CORRECT

**CCB247**

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### *Prediction of Hybridisation; Types of Hybridisation*

**48.** In following hydrocarbon which one has 75% p – character and 25% s – character :
(A) $C_2H_4$ (B) $C_2H_2$ (C) $CH_4$ (D) All of the above

**CCB185**

**49.** The type of hybrid orbitals used by chlorine atom in $ClO^{\ominus}$, $ClO_2^{\ominus}$, $ClO_3^{\ominus}$ and $ClO_4^{\ominus}$ is/are :
(A) sp, $sp^2$, $sp^3$ and $sp^3d$ (B) sp and $sp^3$
(C) Only $sp^3$ (D) Only sp

**CCB187**

**50.** In the series, ethane, ethylene and acetylene, the carbon-hydrogen bond length is :
(A) Equal in all the three (B) Largest in ethane
(C) Smallest in ethylene (D) Largest in acetylene

**CCB188**

**51.** The hybridization of phosphorus in $POCl_3$ is the same as :
(A) P in $\underline{P}Cl_3$ (B) S in $\underline{S}F_4$
(C) Cl in $\underline{Cl}F_3$ (D) B in $\underline{B}Cl_3$

**CCB189**



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**52.** The hybridization state of the central atom in $HgCl_2$ is :
(A) sp (B) $sp^2$ (C) $sp^3$ (D) $dsp^2$

**CCB191**

**53.** Which of the following compounds has four bond pair and one lone pair ?
(A) $NH_4^{\oplus}$ (B) $ICl_4^{\oplus}$ (C) $SF_4$ (D) $XeF_4$

**CCB192**

**54.** Choose the correct option regarding hybridisation of underlined atom :
(A) $\underline{Xe}F_2 \rightarrow sp$ (B) $\underline{Xe}F_4 \rightarrow sp^3$ (C) $\underline{P}OCl_3 \rightarrow sp^3$ (D) $\underline{I}F_5 \rightarrow sp^3d$

**CCB198**

**55.** C–H bond distance is the longest in:
(A) $C_2H_2$ (B) $C_2H_4$ (C) $C_2H_6$ (D) $C_2H_2Br_2$

**CCB244**

## *Calculation of Number of p pi-p pi and p pi-d pi Bonds*

**56.** Which of the following has $p\pi - d\pi$ bonding
(A) $NO_3^{\ominus}$ (B) $CO_3^{2\ominus}$ (C) $BO_3^{3\ominus}$ (D) $SO_3^{2\ominus}$

**CCB179**

**57.** $XeO_4$ molecule is tetrahedral having :
(A) Two $p\pi - d\pi$ bonds (B) Four $p\pi - d\pi$ bonds
(C) One $p\pi - d\pi$ bond (D) Three $p\pi - d\pi$ bonds

**CCB194**

## *Atomic Orbitals Involved in Hybridisation*

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**58.** $SeF_6$ is $sp^3d^2$ hybridised and octahedral in shape, which d – orbitals are involved in hybridisation ?
(A) $d_{x^2-y^2}, d_{xy}$ (B) $d_{x^2-y^2}, d_{z^2}$ (C) $d_{xy}, d_{yz}$ (D) $d_{z^2}, d_{xy}$

**CCB195**

**59.** $IF_7$ is $sp^3d^3$ hybridised and pentagonal bipyramid in shape, which d – orbitals are involved in hybridisation ?
(A) $d_{xy}, d_{yz}, d_{xz}$ (B) $d_{x^2-y^2}, d_{z^2}, d_{xy}$
(C) $d_{x^2-y^2}, d_{yz}, d_{xz}$ (D) $d_{x^2-y^2}, d_{z^2}, d_{yz}$

**CCB196**

## *Solid State Hybridisation*

**60.** What is the difference between bond angles in cationic species of $PCl_5$ and $PBr_5$ in solid state ?
(A) 60° (B) 109°28' (C) 0° (D) 90°

**CCB235**

**61.** What is the hybridisation state of Cl in cationic part of solid $Cl_2O_6$ ?
(A) $sp^3$ (B) sp (C) $sp^2$ (D) $sp^3d$

**CCB251**

## *Introduction of VSEPR Theory*

**62.** Which of the following orders of repulsive interactions is correct ?
(A) lp - lp > lp - bp > bp - bp (B) bp - bp > lp - bp > lp - lp
(C) bp - bp > lp - lp > lp - bp (D) lp - lp > bp - bp > lp - bp

**CCB206**



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*VSEPR Table*

**63.** The shape of $IF_4^{\oplus}$ will be :

(A) Square planar (B) Tetrahedral

(C) Pentagonal bipyramidal (D) See-saw shape

**CCB005**

**64.** The Shape of $SF_4$ is :

(A) Tetrahedral (B) trigonal bipyramidal

(C) See – Saw (D) Octahedral

**CCB207**

**65.** Choose correct matching:

(A) $SO_2 \rightarrow$ Linear (B) $BF_3 \rightarrow$ trigonal planar

(C) $PCl_5 \rightarrow$ octahedral (D) $SF_6 \rightarrow$ tetrahedral

**CCB208**

**66.** The shape of $ClO_3^{\ominus}$ ion according to Valence Shell Electron Pair Repulsion (VSEPR) theory will be :

(A) Triangular planar (B) Pyramidal (C) Tetrahedral (D) Square planar

**CCB209**

**67.** The geometry of $XeO_3F_2$ is :

(A) Triangular planar (B) Trigonal bipyramidal

(C) Square planar (D) Tetrahedral

**CCB210**

**68.** Which of the set of species have same hybridisation state but different shapes ?

(A) $NO_2^{\oplus}, NO_3^{\ominus}, NO_2^{\ominus}$ (B) $ClO_4^{\ominus}$, $SF_4$, $XeF_4$

(C) $NH_4^{\oplus}$, $H_3O^{\oplus}$, $OF_2$ (D) $SO_4^{2\ominus}, PO_4^{3\ominus}, NH_4^{\oplus}$

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**CCB211**

**69.** Which of the following species are expected to be planar?

(a) $NH_3$ (b) $NH_3^{2\oplus}$ (c) $CH_3^{\oplus}$ (d) $PCl_3$

The correct answer is :

(A) b and c (B) c and d (C) b and d (D) a and d

**CCB214**

**70.** Amongst $NO_3^{\ominus}, AsO_3^{3\ominus}, CO_3^{2\ominus}, ClO_3^{\ominus}, SO_3^{2\ominus}$ and $BO_3^{3\ominus}$, In which set all species are non planar :

(A) $CO_3^{2\ominus}, SO_3^{2\ominus}$ and $BO_3^{3\ominus}$ (B) $AsO_3^{3\ominus}, ClO_3^{\ominus}$ and $SO_3^{2\ominus}$

(C) $NO_3^{\ominus}, CO_3^{2\ominus}$ and $BO_3^{3\ominus}$ (D) $SO_3^{2\ominus}, NO_3^{\ominus}$ and $BO_3^{3\ominus}$

**CCB216**

**71.** Which of the following has a geometry different from the other three species (having the same geometry)?

(A) $BF_4^{\ominus}$ (B) $SO_4^{2\ominus}$ (C) $XeF_4$ (D) $PH_4^{\oplus}$

**CCB218**

**72.** In the protonation of $H_2O$, change occurs in :

(A) Hybridisation state of oxygen (B) Shape of molecule

(C) Hybridisation and shape both (D) None

**CCB220**



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**73.** The bond angle at oxygen and hybridization of oxygen in ether ($CH_3OCH_3$) is :

(A) 106°51', $sp^3$ (B) 104°31', $sp^3$ (C) 110°, $sp^3$ (D) 109°28', $sp^3$

**CCB245**

### *Maximum Atoms in A Plane; Number of Such Planes*

**74.** Calculate the maximum number of atoms lying in a plane for $NH_4^{\oplus}$ :

(A) 4 (B) 5 (C) 3 (D) 2

**CCB215**

### *Bond Parameters; Bond Enthalpy*

**75.** Maximum bond energy is in :

(A) $F_2$ (B) $N_2$ (C) $O_2$ (D) equal

**CCB176**

### *Bond Parameters; Bond length*

**76.** Which of the following has been arranged in order of decreasing bond length ?

(A) P–O > Cl–O > S–O (B) P–O > S–O > Cl–O

(C) S–O > Cl–O > P–O (D) Cl–O > S–O > P–O

**CCB246**

### *Bond Parameters; Bond Angle*

**77.** The order of increasing adjacent bond angle in the molecules $BeCl_2$, $BCl_3$, $CCl_4$ and $SF_6$ is :

(A) $SF_6 < CCl_4 < BCl_3 < BeCl_2$ (B) $BeCl_2 < BCl_3 < CCl_4 < SF_6$

(C) $SF_6 < CCl_4 < BeCl_2 < BCl_3$ (D) $BCl_3 < BeCl_2 < SF_6 < CCl_4$

**CCB190**

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**78.** The bond angle in $H_2O$ molecule is less than that of $NH_3$ molecule because :

(A) The hybridisation of O in $H_2O$ and N in $NH_3$ is different

(B) The atomic radii of N and O are different

(C) There is one lone pair of electrons on O and two lone pairs of electrons on N

(D) There are two lone pairs of electrons on O and one lone pair of electrons on N

**CCB212**

### *Concept of Resonance; Bond Order; Comparison of Bond Length and Bond Strength using Bond Order*

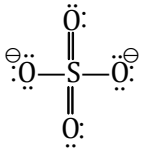
**79.** The weakest Cl –O bond found in:

(A) $ClO_4^{\ominus}$ (B) $ClO_2^{\ominus}$

(C) $ClO^{\ominus}$ (D) $ClO_3^{\ominus}$

**CCB149**

**80.** One of the resonating structure of $SO_4^{2\ominus}$ is :



Which set of average formal charge on oxygen and bond order is correct ?

(A) –0.5 and 1.5 (B) –1.5 and 3

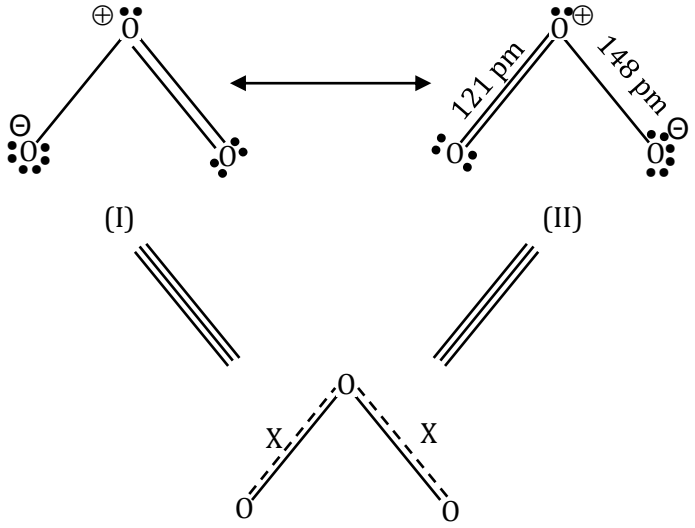
(C) –2 and 3 (D) –1.5 and 1.5

**CCB150**



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**81.** $O_3$ (ozone) molecule can be equally represented by the structures I and II shown below :




Which of the following value of x is correct ?

(A) 148 pm (B) 121 pm

(C) between 121 pm and 148 pm (D) more than 148 pm

**CCB152**

**82.** Point out incorrect statement about resonance

(A) Resonance structures should have equal energy

(B) In resonance structures, the constituent atoms should be in the same position

(C) In resonance structures, there should not be the same number of electron pairs

(D) Resonance structures should differ only in the location of electrons around the constituent atoms

**CCB154**

### *Bent's Rule*

**83.** Choose the INCORRECT statements :

(A) $d_{P-F}$ (axial) > $d_{P-F}$ (equatorial) in $PF_3Cl_2$

(B) $d_{P-F} > d_{P-Cl}$ in $PF_2Cl_3$

(C) $d_{P-Cl}$ ($PF_2Cl_3$) > $d_{P-Cl}$ ($PF_3Cl_2$)

(D) All $d_{P-Cl}$ (in $PF_2Cl_3$) are identical while all $d_{P-F}$ (in $PF_3Cl_2$) are not identical

**CCB236**

**84.** Order of C – H bond lengths :

(A) $d_{C-H}$ ($CH_4$) (B) $d_{C-H}$ ($CH_3F$)

(C) $d_{CH}$ ($CH_2F_2$) (D) $d_{C-H}$ ($CHF_3$)

(A) A > B > C > D (B) D > C > B > A

(C) A > D > B > C (D) D > A > B > C

**CCB237**

**85.** In which of the following molecule C—C bond length will be highest?

(A) $CF_3 - CF_3$ (B) $F_2CH - CHF_2$

(C) $FCH_2 - CH_2F$ (D) $CH_3–CF_3$

**CCB248**



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### *Drago's Rule*

**86.** Among the following, the CORRECT statement is :

(A) Between $NH_3$ and $PH_3$, $NH_3$ is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional

(B) Between $NH_3$ and $PH_3$, $PH_3$ is a better electron donor because the lone pair of electrons occupies $sp^3$ orbital and is more directional

(C) Between $NH_3$ and $PH_3$, $NH_3$ is a better electron donor because the lone pair of electrons occupies $sp^3$ orbital and is more directional

(D) Between $NH_3$ and $PH_3$, $PH_3$ is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional

**CCB240**

**87.** Which of the following order is **CORRECT** for increasing p-character in orbital used for bonding by central atom

(A) $SiH_4 > CH_4$ (B) $H_2S > H_2O$ (C) $PH_4^{\oplus} > PH_3$ (D) $NH_3 > PH_3$

**CCB250**

**88.** The species in which the central atom uses $sp^2$ hybrid orbitals in its bonding is

(A) $PH_3$ (B) $NH_3$ (C) $H_3C^{\oplus}$ (D) $SbH_3$

**CCB252**

### *Odd Electron Molecules*

**89.** An oxide of chlorine which is an odd electron molecule is:

(A) $ClO_3$ (B) $Cl_2O_6$ (C) $Cl_2O_7$ (D) $Cl_2O$

**CCB145**

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**90.** In which of the following processes, the magnetic behaviour of the species is changed ?

(A) $2CH_3 \rightarrow C_2H_6$ (B) $2NO_2 \rightarrow N_2O_4$

(C) $2ClO_3 \rightarrow Cl_2O_6$ (D) All of these

**CCB241**

**91.** Which of the following odd $e^{\ominus}$ species is planar ?

(A) $\dot{C}H_3$ (B) $\dot{C}lO_3$ (C) $\dot{C}F_3$ (D) None of these

**CCB242**

### *Molecules which Do Not Exist*

**92.** Which of the following molecules do not exist ?

(A) $BF_6^{2\ominus}$ (B) $NCl_3$ (C) $NOF_3$ (D) $XeF_5^{\ominus}$

**CCB259**

**93.** Select non existing species :

(A) $PH_3$ (B) $PH_4^{\oplus}$ (C) $[PF_6]^{\ominus}$ (D) None of these

**CCB260**

**94.** Select non existing species

(A) $PF_5$ (B) $SF_6$ (C) $PH_5$ (D) $SF_4$

**CCB261**

**95.** Among the fluorides below, the one which does not exist is -

(A) $XeF_4$ (B) $BeF_4$ (C) $SF_4$ (D) $CF_4$

**CCB262**



For More Join: @IITJEE_Advanced

**96.** Which of the following compound exist -
(A) $XeH_2$ (B) $ClF_7$ (C) $PH_5$ (D) $ICl_3$

**CCB263**

### *Dipole Moment; Calculation of Percent Ionic Character*

**97.** The experimental value of the dipole moment of HCl is 1.03 D. The length of the H – Cl bond is 1.275 Å. The percentage of ionic character in HCl is nearly :
(A) 43 (B) 21 (C) 17 (D) 7

**CCB269**

### *Polarity of Species*

**98.** The dipole moment of $NH_3$ is :
(A) Less than dipole moment of $NCl_3$
(B) Higher than dipole moment of $NCl_3$
(C) Equal to the dipole moment of $NCl_3$
(D) None of these

**CCB264**

**99.** Species having zero dipole moment :
(A) $XeF_4$
(B) 1,2,4 trichloro benzene
(C) $SF_4$
(D) $CH_2Cl_2$

**CCB266**

**100.** The correct order of dipole moment is :
(A) $CH_4 < NF_3 < NH_3 < H_2O$
(B) $NF_3 < CH_4 < NH_3 < H_2O$
(C) $NH_3 < NF_3 < CH_4 < H_2O$
(D) $H_2O < NH_3 < NF_3 < CH_4$

**CCB267**

**101.** Which of the following has the highest value of dipole moment?
(A) HCl (B) HF (C) HI (D) HBr

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**CCB268**

**102.** Which of the following species are polar ?
(a) $C_6H_6$ (b) $XeF_2$ (c) $SO_2$ (d) $SF_4$ (e) $SF_6$
Correct answer is :
(A) (b) and (d)
(B) (a), (b) and (e)
(C) (a) and (e)
(D) (c) and (d)

**CCB270**

**103.** Which contains both polar and non-polar bonds ?
(A) $NH_4Cl$ (B) HCN (C) $H_2O_2$ (D) $CH_4$

**CCB273**

**104.** Find out the incorrect order of the dipole moment among the following pair of compound :
(A) $NH_3 > NF_3$
(B) p-dichloro benzene > o-dichloro benzene
(C) $CH_3Cl > CH_2Cl_2$
(D) $SiF_4 < SF_4$

**CCB274**

**105.** In which of the following both are nonpolar?
(A) $XeF_4$, $O_3$
(B) $XeF_5^{\ominus}$, $N_2O_4$
(C) $SF_4$, $XeF_2$
(D) $BrF_3$, $SCl_2$

**CCB275**



For More Join: @IITJEE_Advanced

**106.** Which of the following molecules is nonpolar ?

(i) $PbCl_4$ (ii) $BF_3$ (iii) $SnCl_2$ (iv) $CS_2$

(A) (i), (ii), (iii) (B) (i), (ii), (iii), (iv) (C) (i), (ii), (iv) (D) (ii), (iii), (iv)

**CCB276**

**107.** Which of the following has the highest dipole moment ?

(A) *o*-Dichlorobenzene (B) *m*-Dichlorobenzene

(C) *p*-Dichlorobenzene (D) All have equal values

**CCB277**

**108.** Both $CO_2$ and $H_2O$ contain polar covalent bonds but $CO_2$ is nonpolar while $H_2O$ is polar because :

(A) H atom is smaller than C atom

(B) $CO_2$ is a linear molecule while $H_2O$ is an angular molecule

(C) O – H bond is more polar than C – H bond

(D) $CO_2$ contains multiple bonds while $H_2O$ has only single bonds

**CCB278**

**109.** $BF_3$ and $NF_3$ both molecules are covalent, but $BF_3$ is non-polar and $NF_3$ is polar. Its reason is

(A) In uncombined state boron is metal and nitrogen is gas

(B) $B-F$ bond has no dipole moment whereas $N-F$ bond has dipole moment

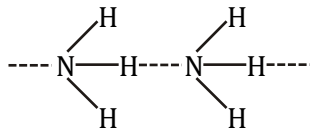
(C) The size of boron atom is smaller than nitrogen

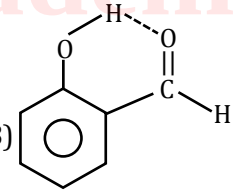
(D) $BF_3$ is planar whereas $NF_3$ is pyramidal

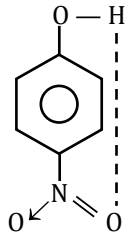
**CCB280**

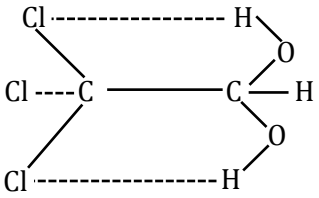
### *Hydrogen Bonding; Strength of Hydrogen Bonding*

**110.** In which of the following molecule e, the shown hydrogen bond is not possible ?

(A) 

(B) 

(C) 

(D) 

**CCB295**

**111.** Maximum number of H–Bond is shown by :

(A) $H_2O$ (B) $H_2Se$ (C) $H_2S$ (D) HF

**CCB296**

**112.** KF combines with HF to form $KHF_2$. The compound contains the species :

(A) $K^{\oplus}$, $F^{\ominus}$ and $H^{\oplus}$ (B) $K^{\oplus}$, $F^{\ominus}$ and HF

(C) $K^{\oplus}$ and $[HF_2]^{\ominus}$ (D) $[KHF]^{\oplus}$ and $F_2$

**CCB298**



For More Join: @IITJEE_Advanced

**113.** For $H_2O_2$, $H_2S$, $H_2O$ and HF, the correct order of decreasing strength of hydrogen bonding is :

(A) $H_2O > HF > H_2O_2 > H_2S$ (B) $HF > H_2O_2 > H_2O > H_2S$

(C) $HF > H_2O > H_2O_2 > H_2S$ (D) $H_2O_2 > H_2O > HF > H_2S$

**CCB307**

**114.** The order of strength of hydrogen bonds is:

(A) NH...N > OH...O > FH...F (B) NH...N < OH...O < FH...F

(C) NH...N > OH...O < FH...F (D) NH...N < OH...O > FH...F

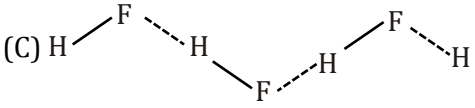
**CCB309**

**115.** The *H* bond in solid *HF* can be best represented as:

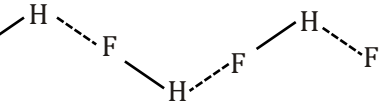
(A) H – F....H – F....H – F (B)



(C)



(D)



**CCB310**

**116.** In which of the following compounds does hydrogen bonding occur

(A) $SiH_4$ (B) LiH (C) HI (D) $NH_3$

**CCB312**

### *Types of Hydrogen Bonding*

**117.** The crystal lattice of ice is mostly formed by :

(A) Ionic forces (B) Covalent bonds

(C) Inter molecular H–bonds (D) Covalent as well as H–bonds

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**CCB297**

**118.** Which of the following compounds would have significant intermolecular hydrogen bonding?

HF, $CH_3OH$, $N_2O_4$, $CH_4$

(A) HF, $N_2O_4$ (B) HF, $CH_4$, $CH_3OH$ (C) HF, $CH_3OH$ (D) $CH_3OH$, $CH_4$

**CCB306**

**119.** Which one of the following does not have intermolecular H-bonding?

(A) $H_2O$ (B) *o*-nitro phenol

(C) HF (D) $CH_3COOH$

**CCB308**

### *Applications of Hydrogen Bonding; Viscosity ; Boiling Point; Volatility; Solubility in Water*

**120.** The correct order of volatility is :

(A) $NH_3 < H_2O$ (B) p – nitro phenol < o – nitro phenol

(C) $CH_3OH > CH_3 – O – CH_3$ (D) HF > HCl

**CCB299**

**121.** Which of the following hydrides has the lowest boiling point ?

(A) $H_2O$ (B) $H_2S$ (C) $H_2Se$ (D) $H_2Te$

**CCB300**

**122.** Ethanol has a higher boiling point than dimethyl ether though they have the same molecular weight. This is due to :

(A) resonance (B) coordinate bonding

(C) hydrogen bonding (D) ionic bonding

**CCB305**



For More Join: @IITJEE_Advanced

**123.** In the following which has highest boiling point
(A) HI (B) HF (C) HBr (D) HCl

**CCB311**

**124.** Acetic acid exists as dimer in benzene due to
(A) Condensation reaction
(B) Hydrogen bonding
(C) Presence of carboxyl group
(D) Presence of hydrogen atom at α- carbon

**CCB313**

**125.** The high density of water compared to ice is due to
(A) Hydrogen bonding interactions
(B) Dipole-dipole interactions
(C) Dipole-induced dipole interactions
(D) Instantaneous dipole-induced dipole interactions

**CCB314**

### *Inert Pair Effect*

**126.** E represents an element belonging to carbon family.
$E + 2X_2 \longrightarrow EX_4$ (X = F, Cl, Br, I)
(A) Stability of $EX_4$ decreases down the $14^{th}$ group
(B) $PbI_4$ does not exist
(C) Ge & Pb forms $EX_2$ as well.
(D) All are correct

**CCB521**

### *Back Bonding*

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**127.** Which of the following statements is not correct ?
(A) $(CH_3)_3COH$ is less acidic than $(CH_3)_3SiOH$.
(B) Like $CO_2$ its analogue of Si is not stable.
(C) In phosgene, C–O bond length is longer than expected while C–Cl bond length is shorter.
(D) None of the above.

**CCB485**

**128.** In $B_3N_3H_6$ compound, the correct option is (State 'T' for True and 'F' for False) :
(P) '3' pπ– pπ bond is present.
(Q) It can show back bonding.
(R) It is non–polar compound with non–polar bonds.
(S) More e– density is present at Boron atom.
(A) TTFF (B) TTTT (C) TTTF (D) FTTF

**CCB486**

**129.** In which of the following cases, the increase in bond angle is not due to back bonding ?
(A) $\hat{C}OC$ angle in $CH_3OCH_3$
(B) $Si\hat{N}Si$ angle in $N(SiH_3)_3$
(C) $F\hat{B}F$ angle in $BF_3$
(D) $F-\hat{P}-F$ angle in $PF_3$

**CCB487**

**130.** Which of the following molecule has large extent of back bonding ?
(A) $O(SiH_3)_2$
(B) $OCl_2$
(C) Both have same extent of back bonding
(D) None of the above

**CCB488**



For More Join: @IITJEE_Advanced

**131.** Choose the incorrect statement from the following.
(A) Bond angles are not affected in $BF_3$ due to back bonding.
(B) Bond angles are affected in $PF_3$ due to back bonding.
(C) Bond angles are not affected in $B(OMe)_3$ due to back bonding.
(D) None of the above

**CCB489**

***Bridge Bonding(3C-2e- , 3C-4e-) , $B_2H_6$ , $Al_2Cl_6$ , Bond Behaviour of $B_2H_6$ ; Bridge Bonding in other Compounds; Reactions of Diborane***

**132.** Which of the following statement is INCORRECT ?
(A) Hybridization state of 'Be' in $BeH_2$ (solid) is $Sp^3$.
(B) $BeH_2$ (solid) is a planar molecule
(C) $BeH_2$ (solid) is a electron deficient molecule.
(D) In $BeH_2$ (solid) $3c–2e^{\ominus}$ bond present

**CCB522**

**133.** Unsymmetrical cleavage of $B_2H_6$ occurs on reaction with :
(A) $Me_3N$: (B) $Me_3P$: (C) $Et_2\ddot{S}$: (D) $H_3N$:

**CCB491**

**134.** The state of hybridization of central atom in dimer form of both $BH_3$ and $BeH_2$ is :
(A) $sp^2$, sp (B) $sp^3$, $sp^2$ (C) $sp^3$, $sp^3$ (D) $sp^2$, $sp^3$

**CCB492**

**135.** Which of the following molecule has complete octet ?
(A) $B_2H_6$ (B) $Al_2Cl_6$ (C) $Be_2Cl_4$ (D) $BeH_2$

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**CCB493**

***Oxyacids of Phosphorus***

**136.** Which of the following oxyacid contains P – P bond?
(A) $H_4P_2O_8$ (B) $H_4P_2O_5$ (C) $H_4P_2O_7$ (D) $H_4P_2O_6$

**CCB523**

**137.** How many of the following molecules have maximum number of P–O–P linkage?
(A) $H_5P_3O_{10}$ (B) $H_5P_5O_{15}$ (C) $H_7P_5O_{16}$ (D) $H_4P_2O_7$

**CCB495**

**138.** The molecular formula of sodium hydrogen phosphate is :
(A) $NaH_2PO_4$ (B) $Na_2HPO_4$ (C) $NaH_2PO_3$ (D) $Na_2HPO_3$

**CCB496**

***Oxyacids of Sulphur***

**139.** The compound having S—S linkage is :
(A) $H_2S_2O_7$ (B) $H_2S_2O_8$ (C) $H_2S_2O_6$ (D) $(SO_3)_3$

**CCB497**

***Oxyacids of Chlorine***

**140.** Cl=O bond is NOT present in:
(A) Chloric acid (B) Hypochlorous acid (C) Chlorous acid (D) Perchloric acid

**CCB524**

**141.** Which of the following compound does not have peroxy linkage?
(A) Marshall's acid (B) Caro's acid (C) Perchloric acid (D) Pernitric acid

**CCB498**



For More Join: @IITJEE_Advanced

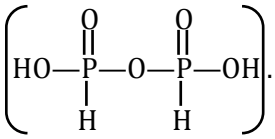
*Hydrolysis*

**142.** $CCl_4$ is inert towards hydrolysis but $SiCl_4$ is readily hydrolysed because
(A) carbon cannot expand its octet but silicon can expand its octet
(B) ionisation potential of carbon is higher than silicon
(C) carbon forms double and triple bonds
(D) electronegativity of carbon is higher than that of silicon

**CCB525**

**143.** Which one of the following compounds gives methane on treatment with water :-
(A) $Al_4C_3$ (B) $CaC_2$ (C) $Mg_2C_3$ (D) SiC

**CCB526**

**144.** Find the product(s) of hydrolysis of pyrophosphorous acid  .
(A) $2H_3PO_4$ (B) $H_3PO_4 + H_3PO_3$ (C) $2H_3PO_3$ (D) $H_3PO_3 + H_3PO_2$

**CCB499**

**145.** Type of hybridization of transition state of $SeF_4$, when it undergoes hydrolysis is :
(A) $sp^3d$ (B) $sp^3d^2$ (C) $dsp^3$ (D) $sp^3$

**CCB500**

**146.** $SbCl_3$ and $BiCl_3$ on hydrolysis gives :
(A) $Sb^{+3}$ and $Bi^{+3}$ (B) $Sb(OH)_3$ and $Bi(OH)_3$
(C) SbOCl and BiOCl (D) None of these

**CCB501**

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**147.** Calculate the moles of oxyacid formed when 18 grm of water reacts with $NCl_3$ and undergoes complete hydrolysis.
(A) 1 mole (B) 3 mole
(C) $\frac{1}{3}$ mole (D) $\frac{1}{2}$ mole

**CCB502**

**148.** Which of the following statements is correct ?
(A) On hydrolysis, $SeF_6$ and $TeF_6$ four water molecules are consumed.
(B) On hydrolysis of $PCl_3$, product obtained has basicity 3.
(C) On hydrolysis of $SF_4$, hybridisation of S in transition state is $sp^3d$.
(D) All molecules of $BF_3$ are not hydrolysed in water.

**CCB503**

**149.** What is the oxidation number of P in product when $POCl_3$ is completely hydorlysed?
(A) +5 (B) +3 (C) +4 (D) +2

**CCB504**

*Introduction of Weak Forces; Types of Weak Forces; Factors Affecting London Dispersion Forces; Comparison of Boiling Point of Species*

**150.** Which is the weakest among the following types of bonds?
(A) Debye force (B) Metallic bond
(C) Dipole-dipole interaction (D) Hydrogen bond

**CCB319**



For More Join: @IITJEE_Advanced

**151.** Which of the following inert gas liquefied easily as compare to others ?
(A) Kr (B) He (C) Ne (D) Ar

**CCB322**

**152.** The relative strength of interionic/intermolecular forces in decreasing order is :
(A) Ion-dipole > Ion-ion > Dipole-dipole (B) Dipole-dipole > Ion-dipole > Ion-ion
(C) Ion-dipole > Dipole-dipole > Ion-ion (D) Ion-ion > Ion-dipole > Dipole-dipole

**CCB323**

**153.** Which of the following boiling point order is correct?
(A) He > $T_2$ > $D_2$ (B) He < $T_2$ < $D_2$ (C) $T_2$ > He > $D_2$ (D) He < $D_2$ < $T_2$

**CCB324**

**154.** Identify the incorrect order of boiling point in the following pairs :
(A) $B(OH)_3 < B(OCH_3)_3$ (B) $NF_3 < N(CH_3)_3$
(C) $BF_3 < B(CH_3)_3$ (D) $C_2H_6 < C_2F_6$

**CCB325**

**155.** Select the CORRECT statement :
(A) Boiling point of $NF_3$ is greater than $NMe_3$
(B) Greater the dipole moment in molecule, greater will be the dipole-dipole interaction between the molecules.
(C) London dispersion force increases with decreasing number of electrons
(D) Boiling point of hydrides of carbon family decreases down the group.

**CCB327**

**156.** In solid argon, the atoms are held together by
(A) Ionic bonds (B) Hydrogen bonds
(C) Van Der Waals forces (D) Hydrophobic forces

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**CCB329**

**157.** Which of the following exhibits the weakest intermolecular forces
(A) He (B) HCl (C) $NH_3$ (D) $H_2O$

**CCB331**

**158.** The intermolecular interaction that is dependent on the inverse cube of distance between the molecules is :
(A) London force (B) Hydrogen bond
(C) Ion-ion interaction (D) Ion-dipole interaction

**CCB333**

**159.** Which one has the highest boiling point ?
(A) Kr (B) Xe
(C) He (D) Ne

**CCB334**

***Molecular Orbital Theory; Drawbacks of Valence Bond Theory; Introduction of Molecular Orbital Theory***

**160.** During the formation of a molecular orbital from atomic orbitals, probability of electron density is
(A) minimum in the nodal plane (B) maximum in the nodal plane
(C) zero in the nodal plane (D) zero on the surface of the lobe

**CCB351**



For More Join: @IITJEE_Advanced

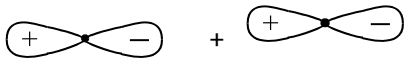
**161.** Out of the following which has smallest bond length

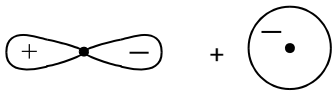
(A) $O_2$ (B) $O_2^{\oplus}$ (C) $O_2^{\ominus}$ (D) $O_2^{2\ominus}$

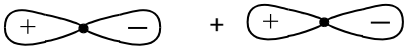
**CCB364**

### *Linear Combination of Atomic Orbitals*

**162.** Which of the following combination of orbitals is CORRECT ?

(A) 

(B) 

(C) 

(D) 

**CCB344**

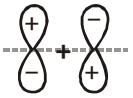
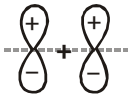
**163.** If the z-axis is taken as the internuclear axis, then which of the following combinations of atomic orbitals is a nonbonding combination?

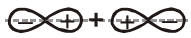
(A) s and $p_y$ (B) $p_x$ and s

(C) $p_x$ and $p_y$ (D) All of these

**CCB345**

**164.** Which of the following overlapping indicates formation of bonding molecular orbitals having gerade symmetry?

(A)  (B) 

(C)  (D) All of these

**CCB352**

**165.** Which of the following molecular orbitals has two different nodal planes

(A) $\sigma_{2s}$ (B) $\pi_{2p_y}$ (C) $\pi^*_{2p_y}$ (D) $\sigma^*_{2p_x}$

**CCB358**

### *Bonding in some homonuclear diatomic molecules (Molecular Orbital Diagram of $H_2$, $He_2$, $Li_2$, $O_2$ ,$B_2$,$C_2$,$N_2$,$F_2$ , $Ne_2$)*

**166.** Which of the following does not exist ?

(A) $B_2$ (B) $C_2$ (C) $He_2$ (D) $N_2$

**CCB343**

**167.** In the formation of $N_2^+$, the electron is removed from :

(A) σ- orbital (B) π -orbital (C) σ* - orbital (D) π*-orbital

**CCB350**

**168.** Which one of the following species is diamagnetic in nature

(A) $He_2^{\oplus}$ (B) $H_2$ (C) $H_2^{\oplus}$ (D) $H_2^{\ominus}$

**CCB357**



For More Join: @IITJEE_Advanced

**169.** LUMO in $N_2$ is :

(A) $\pi^*_{2px}$ (B) $\sigma^*_{2pz}$ (C) $\pi_{2px}$ (D) $\sigma_{2pz}$

**CCB372**

**170.** HOMO in $O_2$ is :

(A) $\pi^*_{2p_x}$ (B) $\sigma^*_{2p_z}$ (C) $\pi_{2p_x}$ (D) $\sigma_{2p_z}$

**CCB373**

**171.** Most stable species is :

(A) $O_2^{1\oplus}$ (B) $O_2^{1\ominus}$

(C) $O_2^{2\ominus}$ (D) All equal

**CCB374**

**172.** The bond energy order of $He_2^{\oplus}$ and $HeH^{\oplus}$ is

(A) $He_2^{\oplus} > HeH^{\oplus}$ (B) $HeH^{\oplus} = He_2^{\oplus}$

(C) $HeH^{\oplus} > He_2^{\oplus}$ (D) Can't be predicted

**CCB376**

**173.** Number of anti bonding electrons in $N_2$ is :

(A) 4 (B) 10 (C) 12 (D) 14

**CCB378**

***Applications of MOT(Molecular Behaviour) ; 1 Bond Order; 2 Bond Length; 3 Bond Strength; 4 Ionisation Energy; 5 Colour***

**174.** Which of the following species will have the minimum bond energy :

(A) $N_2$ (B) $N_2^{\ominus}$ (C) $N_2^{\oplus}$ (D) $N_2^{2\ominus}$

**CCB009**

**175.** Which of the following species will have the minimum bond energy ?

(A) $N_2$ (B) $N_2^{\ominus}$ (C) $N_2^{\oplus}$ (D) $N_2^{2\ominus}$

**CCB368**

**176.** When $N_2$ is ionised to $N_2^{\oplus}$ bond length ..... & if $O_2$ is ionised to $O_2^{\oplus}$ bond length .....

Select the CORRECT option to fill the blank space respectively:

(A) Increases and decreases (B) Decreases and increases

(C) Increases and increases (D) Decreases and decreases

**CCB369**

**177.** Which of the following order of ionisation energy is correct?

(A) $F_2 > F$ (B) $F_2 < F$ (C) $F_2 = F$ (D) Cannot predict

**CCB370**

**178.** Choose Correct order of ionisation energy ?

(A) $O_2 > O$ (B) $O_2 < O$ (C) $O_2 = O$ (D) Cannot predict

**CCB371**

**179.** The highest bond length is present in :

(A) $B_2$ (B) $B_2^{2\ominus}$ (C) $B_2^{1\oplus}$ (D) All equal

**CCB375**



For More Join: @IITJEE_Advanced

***Concept of Hydration; Hydration Energy; Hydrated Radius; Ionic Mobility in Water; Conductivity in Water; Condition for Solubility of Ionic Compounds in Water***

**180.** Hydrated radius of $M^{\oplus}$ ions in decreasing order :
(A) $Li^{\oplus}(aq) > Na^{\oplus}(aq) > K^{\oplus}(aq) > Rb^{\oplus}(aq) > Cs^{\oplus}(aq)$
(B) $Li^{\oplus}(aq) > Na^{\oplus}(aq) < K^{\oplus}(aq) < Rb^{\oplus}(aq) < Cs^{\oplus}(aq)$
(C) $Li^{\oplus}(aq) > Na^{\oplus}(aq) > K^{\oplus}(aq) > Rb^{\oplus}(aq) < Cs^{\oplus}(aq)$
(D) $Li^{\oplus}(aq) < Na^{\oplus}(aq) < K^{\oplus}(aq) < Rb^{\oplus}(aq) < Cs^{\oplus}(aq)$

**CCB002**

**181.** The hydration energy of $Mg^{2\oplus}$ is greater than the hydration energy of :
(A) $Al^{3\oplus}$ (B) $Mg^{3\oplus}$
(C) $Na^{\oplus}$ (D)$Be^{2\oplus}$

**CCB388**

**182.** Which of the compound is least soluble in water ?
(A) AgF (B) AgCl
(C) AgBr (D) AgI

**CCB389**

**183.** Ionic conductance of hydrated $M^{\oplus}$ ions are in the order :
(A) $Li^{\oplus}(aq) > Na^{\oplus}(aq) > K^{\oplus}(aq) > Rb^{\oplus}(aq) > Cs^{\oplus}(aq)$
(B) $Li^{\oplus}(aq) > Na^{\oplus}(aq) < K^{\oplus}(aq) < Rb^{\oplus}(aq) < Cs^{\oplus}(aq)$
(C) $Li^{\oplus}(aq) > Na^{\oplus}(aq) > K^{\oplus}(aq) > Rb^{\oplus}(aq) < Cs^{\oplus}(aq)$
(D) $Li^{\oplus}(aq) < Na^{\oplus}(aq) < K^{\oplus}(aq) < Rb^{\oplus}(aq) < Cs^{\oplus}(aq)$

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**CCB390**

**184.** The ionic mobility of alkali metal ions in aqueous solution is maximum for
(A) $Li^{\oplus}$ (B) $Na^{\oplus}$ (C) $K^{\oplus}$ (D) $Rb^{\oplus}$

**CCB391**

**185.** Sodium sulphate is soluble in water whereas barium sulphate is sparingly soluble because :
(a) The hydration energy of $Na_2SO_4$ is less than its lattice energy
(b) The hydration energy of $Na_2SO_4$ is more than its lattice energy
(c) The lattice energy of $BaSO_4$ is more than its hydration energy
(d) The lattice energy has no role to play in solubility
(A) a & b (B) b & c
(C) c & d (D) b & d

**CCB392**

**186.** Choose the compound from each of the following pairs that has the more solubility in water :-
(a) $BeSO_4$, $BaSO_4$ (b) LiOH, CsOH (c) AgCl, AgI
(A) $BeSO_4$, CsOH, AgI (B) $BeSO_4$, CsOH, AgCl
(C) $BaSO_4$, LiOH, AgCl (D) $BaSO_4$, LiOH, AgI

**CCB393**

**187.** Which of the following metal ion will have maximum hydrated radius :
(A) $Be^{2\oplus}$ (B) $Mg^{2\oplus}$
(C) $Ca^{2\oplus}$ (D) $Sr^{2\oplus}$

**CCB396**



For More Join: @IITJEE_Advanced

### *Fajan's Rule; Polarisation; Polarising Power; Polarizability; Factors Affecting Polarisation*

**188.** The charge/size ratio of a cation determines its polarizing power. Which one of the following sequences represents the increasing order of the polarizing power of the cationic species, $K^{\oplus}$, $Ca^{2\oplus}$, $Mg^{2\oplus}$, $Be^{2\oplus}$ ?

(A) $Be^{2\oplus} < K^{\oplus} < Ca^{2\oplus} < Mg^{2\oplus}$ (B) $K^{\oplus} < Ca^{2\oplus} < Mg^{2\oplus} < Be^{2\oplus}$

(C) $Ca^{2\oplus} < Mg^{2\oplus} < Be^{2\oplus} < K^{\oplus}$ (D) $Mg^{2\oplus} < Be^{2\oplus} < K^{\oplus} < Ca^{2\oplus}$

**CCB399**

**189.** $CCl_4$ is more covalent than LiCl because :

(A) There is more polarization of Cl in $CCl_4$ (B) There is more polarization of Cl in LiCl

(C) $CCl_4$ has more weight (D) None of above

**CCB400**

**190.** Polarization is the distortion of the shape of an anion by an adjacently placed cation. Which of the following statements is correct

(A) Maximum polarization is brought about by a cation of high charge

(B) Minimum polarization is brought about by a cation of low radius

(C) A large cation is likely to bring about a large degree of polarization

(D) A small anion is likely to undergo a large degree of polarization

**CCB408**

**191.** In which of the following molecule is the covalent bond most polar

(A) HI (B) HBr (C) HCl (D) $H_2$

**CCB409**

**192.** Which of the following has a high polarising power

(A) $Mg^{2\oplus}$ (B) $Al^{3\oplus}$ (C) $Na^{\oplus}$ (D) $Ca^{2\oplus}$

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**CCB411**

**193.** According to Fajan's rule, covalent bond is favoured by

(A) Large cation and small anion (B) Large cation and large anion

(C) Small cation and large anion (D) Small cation and small anion

**CCB413**

### *Properties Affected by Polarisation; Covalent Character of Ionic Compounds*

**194.** Amongst LiCl, RbCl, $BeCl_2$ and $MgCl_2$ the compounds with the greatest and the least ionic character, respectively are :

(A) LiCl and RbCl (B) RbCl and $BeCl_2$

(C) RbCl and $MgCl_2$ (D) $MgCl_2$ and $BeCl_2$

**CCB397**

**195.** Compound with maximum ionic character is formed from :

(A) Na and Cl (B) Cs and F

(C) Cs and I (D) Na and F

**CCB398**

**196.** Select correct order out of given options :

(A) $BeCO_3 < BaCO_3$ : Covalent character (B) BeO > SrO : lattice energy

(C) $Be^{2\oplus} < Li^{\oplus}$ : Hydration energy (D) $Be^{2\oplus}$ (aq.) > $Li^{\oplus}$ (aq.) : Ionic mobility

**CCB402**



For More Join: @IITJEE_Advanced

**197.** Maximum covalent character is associated with the compound

(A) NaI (B) $MgI_2$ (C) $AlCl_3$ (D) $All_3$

**CCB412**

### *Properties Affected by Polarisation; Melting Point of Ionic Compounds*

**198.** Which of the following has highest melting point

(A) $BeCl_2$ (B) $MgCl_2$ (C) $CaCl_2$ (D) $BaCl_2$

**CCB405**

**199.** Which compound is highest covalent

(A) LiCl (B) LiF (C) LiBr (D) LiI

**CCB407**

### *Properties Affected by Polarisation; Solubility of Ionic Compounds in Water*

**200.** Which of the following set of characteristics lead to the increase in solubility of ionic substances?

(A) High dipole moment, strong attraction by an ion towards solvent and large solvation energy

(B) Low dipole moment, weak attraction by an ion towards solvent and high solvation energy

(C) High dipole moment, strong attraction by an ion towards solvent and low solvation energy

(D) High dipole moment, weak attraction by an ion towards solvent and large solvation energy

**CCB403**

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# EXERCISE (O-2)

## MULTIPLE CORRECT TYPE QUESTIONS

### *Kossel -Lewis approach to Chemical Bonding; Lewis Dot Structure*

**1.** Which of the following compounds contain ionic, covalent and co-ordinate bonds?

(A) $NH_4Cl$ (B) KCN (C) $NaBF_4$ (D) NaOH

**CCB135**

### *Prediction of Hybridisation; Types of Hybridisation*

**2.** Which of the following statement(s) is / are not correct?

(A) Hybridization is the mixing of atomic orbitals of large energy difference.

(B) $sp^2$–hybrid orbitals are formed from two *p* - atomic orbitals and one *s*- atomic orbital

(C) $sp^3d^2$–hybrid orbitals are all at 90° to one another

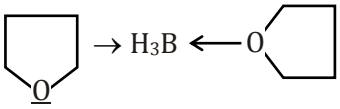
(D) $sp^3$–hybrid orbitals are directed towards the corners of a regular tetrahedron

**CCB201**

**3.** In which of the following process(s) hybridisation of underlined atom does not change ?

(A) $\underline{N}H_3 + BF_3 \rightarrow H_3N.BF_3$

(B) $\underline{S}iF_4 + 2F^{\ominus} \rightarrow [SiF_6]^{2\ominus}$

(C) $BH_3 +$  $\rightarrow H_3B \leftarrow$ 

(D) $H_3\underline{B}O_3 + \overset{\ominus}{\underset{\cdot\cdot}{O}}H \rightarrow [B(OH)_4]^{\ominus}$

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**CCB223**

### *Atomic Orbitals Involved in Hybridisation*

**4.** Which of the following 'd' orbital(s) participate in the hybridistaion for Xe in the cationic part of $XeF_6(s)$ :-

(A) $d_{xy}$ (B) $d_{x^2-y^2}$ (C) $d_{z^2}$ (D) $d_{y^2}$

**CCB283**

### *VSEPR Table*

**5.** Which of the following species are linear ?

(A) $ICl_2^{\ominus}$ (B) $I_3^{\ominus}$ (C) $N_3^{\ominus}$ (D) $ClO_2^{\oplus}$

**CCB226**

**6.** Which of the following molecules have bent shape.

Where A is central atom, B is surrounding atom and E is lone pair -

(A) $AB_2E_2$ (B) $AB_2E$ (C) $AB_3E$ (D) $AB_2$

**CCB227**

### *Bond Parameters; Bond Angle*

**7.** INCORRECT order of bond angle is :

(A) $OCl_2 > SF_2$ (B) $H_2O > OF_2$

(C) $SO_4^{2\ominus} > CF_4$ (D) $NF_3 > NH_3$

**CCB224**



For More Join: @IITJEE_Advanced

### *Concept of Resonance; Bond Order; Comparison of Bond Length and Bond Strength using Bond Order*

**8.** Select the correct statement -

(A) C–O bond length of CO molecule is higher than the C–O bond length of $CO_2$

(B) C–O bond length of CO molecule is lower than the C–O bond length of $CO_2$

(C) N–O bond length of $NO_3^{\ominus}$ ion is higher than the N–O bond length of $NO_2^{\ominus}$

(D) N–O bond length of $NO_3^{\ominus}$ ion is lower than the N–O bond length of $NO_2^{\ominus}$

**CCB155**

**9.** $CO_3^{2\ominus}$ anion has which of the following characteristics

(A) Bonds of unequal length
(B) $sp^2$ hybridization of *C* atom
(C) Resonance stabilization
(D) Same bond angles

**CCB202**

**10.** Select the CORRECT statement(s) :

(A) Bond angle order : $CH_4 > CF_4$

(B) Bond length order : $d_{N-O}(NO_2^{\ominus}) < d_{N-O}(NO_3^{\ominus})$

(C) Bond order of S – O : $SO_4^{2\ominus} > SO_3^{2\ominus}$

(D) Bond angle order : $NH_3 < PH_3$

**CCB225**

### *Drago's Rule*

**11.** In which of the following, the theory of hybridisation does not help to predict the bond angle ?

(A) $PH_3$ (B) $SbH_3$ (C) $SiH_4$ (D) $H_2S$

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**CCB282**

### *Odd Electron Molecules*

**12.** Choose the correct options for $NO_2$ :

(A) It is odd electron specie
(B) N–O bond order = 1.5
(C) Paramagnetic specie
(D) Isoelectronic with $CO_2$

**CCB254**

**13.** The number of specie(s) which are not perfectly planar.

(A) $\dot{C}H_3$ (B) $\dot{C}F_3$ (C) $CHF_3$ (D) $CH_2F_2$

**CCB255**

### *Molecules which Do Not Exist*

**14.** Which of the following molecules do not exist ?

(A) $Br_2O$ (B) $SF_6$ (C) $NCl_5$ (D) $OF_4$

**CCB284**

**15.** Which of the following molecule exist-

(A) $SF_6$ (B) $IH_3$ (C) $SH_2$ (D) $PCl_5$

**CCB286**

**16.** Which of the following molecules does not exist and non-existence reason is same as that of $SH_6$

(A) $OF_4$ (B) $SH_4$ (C) $PBr_6^{\ominus}$ (D) $XeH_4$

**CCB287**



For More Join: @IITJEE_Advanced

### *Polarity of Species*

**17.** Which of the following compounds possesses zero dipole moment?

(A) Water (B) Benzene

(C) Carbon tetrachloride (D) Boron trifluoride

**CCB288**

### *Types of Hydrogen Bonding*

**18.** In which of the following chemical species hydrogen bonding takes places -

(A) $Na_2HPO_3(s)$ (B) $K_2HPO_4(s)$ (C) $KHCO_3(s)$ (D) $NaHCO_3(s)$

**CCB317**

### *Applications of Hydrogen Bonding; Viscosity ; Boiling Point; Volatility; Solubility in Water*

**19.** Which of the following statements are correct ?

(A) The crystal lattice of ice is formed by covalent as well as hydrogen bonds

(B) The density of water increases when heated from 0°C to 4°C

(C) Above 4°C the thermal agitation of water molecules increases. Therefore, intermolecular distance increases and water starts expanding

(D) The density of water decreases from 0°C to a maximum at 4° C

**CCB315**

### *Inert Pair Effect*

**20.** Which of following molecules acts as good oxidizing agent ?

(A) $NaBiO_3$ (B) $PbO_2$ (C) $TlCl_3$ (D) $PbCl_2$

**CCB527**

**21.** Which of the following is CORRECT order of stability ?

(A) $Pb^{2\oplus} > Pb^{4\oplus}$ (B) $Bi^{3\oplus} > Bi^{5\oplus}$ (C) $Tl^{\oplus} < Tl^{3\oplus}$ (D) $Sn^{2\oplus} > Sn^{4\oplus}$

**CCB528**

### *Back Bonding*

**22.** Which of the following molecule(s) is/are having $p\pi-p\pi$ back bonding ?

(A)$BF_3$ (B) $BCl_3$ (C) $B_3N_3H_6$ (D) None of these

**CCB529**

**23.** Which of the following statement is/are correct for $CCl_3^-$ and $CCl_2$ ?

(A) Back bonding in both cases from Cl to C-atom

(B) Back bonding in both cases from C to Cl-atom

(C) Back bonding in $CCl_3^-$ is from C to Cl but reverse in $CCl_2$.

(D) Direction of back bonding is just opposite in two cases.

**CCB530**

### *Bridge Bonding(3C-2e- , 3C-4e-) , $B_2H_6$ , $Al_2Cl_6$ , Bond Behaviour of $B_2H_6$ ; Bridge Bonding in other Compounds; Reactions of Diborane*

**24.** Which of following are correct.

(A) $B_2H_6$ is non-planar.

(B) $B_2H_6$ undergo symmetrical cleavage with $PF_3$ , CO and $(C_2H_5)_3N$

(C) $B_2H_6$ undergo unsymmetrical cleavage with $NH_3$, $CH_3NH_2$, $(CH_3)_2NH$.

(D) $BeH_2(s)$ , $Ga_2Me_6$ , $Al_2Me_6$ have two type of bonds (2C–2e bond as well as 3C–2e bond)

**CCB531**



For More Join: @IITJEE_Advanced

**25.** Which of the following molecule has incomplete octet?

(A) $B_2H_6$ (B) $Al_2Cl_6$

(C) $Be_2Cl_4$ (D) $Be_2H_4$ (vapor)

**CCB532**

### *Oxyacids of Sulphur*

**26.** The acid having two directly connected sulphur atoms not present :

(A) $H_2S_2O_7$ (B) $H_2S_2O_8$ (C) $H_2S_2O_6$ (D) $H_2SO_5$

**CCB533**

**27.** Which among the following is/are peroxo acid(s) ?

(A) $H_2S_2O_3$ (B) $H_2SO_5$

(C) $H_2S_2O_7$ (D) $H_2S_2O_8$

**CCB534**

### *Hydrolysis*

**28.** When $PCl_3$ undergoes in hydrolysis then X oxyacid of phosphorus is formed then Select the correct statement about X :

(A) Two P – OH bond (B) One P – H bond

(C) One P = O bond (D) Three P – OH bond

**CCB535**

### *Introduction of Weak Forces; Types of Weak Forces; Factors Affecting London Dispersion Forces; Comparison of Boiling Point of Species*

**29.** Which of the following factors are responsible for origination of Van Der Waals forces ?

(A) Instantaneous dipole-induced dipole interaction

(B) Dipole-induced dipole interaction

(C) Dipole-dipole interaction

(D) Size of molecule

**CCB335**

**30.** Which of the following are true?

(A) Van Der Waals forces are responsible for the formation of molecular crystals

(B) Branching lowers the boiling points of isomeric organic compounds due to decrease in Van Der Waals forces of attraction

(C) In graphite, Van Der Waals forces act between the carbon layers

(D) In diamond, Van Der Waals forces act between the carbon layers

**CCB336**

**31.** London forces depends upon :

(A) Molecular weight (B) Number of polarisable electrons

(C) Molecular size (D) None of these

**CCB337**

### *Linear Combination of Atomic Orbitals*

**32.** Which of the following overlapping results non bonding molecular orbital (if overlapping axis is x-axis) ?

(A) s of A & $p_y$ of B (B) $p_z$ of A & $p_z$of B

(C) $p_y$ of A & $p_z$ of B (D) $d_{xy}$ of A & s of B

**CCB359**



For More Join: @IITJEE_Advanced

***Bonding in some homonuclear diatomic molecules (Molecular Orbital Diagram of $H_2$, $He_2$, $Li_2$, $O_2$, $B_2$, $C_2$, $N_2$, $F_2$ , $Ne_2$)***

**33.** Which of the following statements are true for these given species : $N_2$, CO, $CN^{\ominus}$ and $NO^{\oplus}$.

(A) All species are paramagnetic (B) The species are isoelectronic

(C) All the species have dipole moment (D) All the species are linear

**CCB360**

**34.** Which of the following have unpaired electron(s)

(A) $O_2^{\oplus}$ (B) $O_2^{\ominus}$ (C) NO (D) $H_2^{\oplus}$

**CCB361**

**35.** Which of the following is/are paramagnetic ?

(A) $B_2$ (B) $O_2$ (C) $N_2$ (D) $He_2$

**CCB362**

**36.** Which of the following is/are correct?

(A) During $N_2^{\oplus}$ formation, one electron is removed from the bonding molecular orbitals

(B) During $O_2^{\oplus}$ formation, one electron is removed from the antibonding molecular orbitals

(C) During $O_2^{\ominus}$ formation, one electron is added to the bonding molecular orbitals

(D) During $N_2^{\ominus}$ formation, one electron is added to the bonding molecular orbitals

**CCB363**

**37.** Which of the following statement is/are correct

(A) The peroxide ion has a bond order of 1 while the oxygen molecule has a bond order of 2

(B) The peroxide ion has a weaker bond than the dioxygen molecule has.

(C) The peroxide ion as well as the dioxygen molecules are paramagnetic

(D) The bond length of the peroxide ion is greater than that of the dioxygen molecule

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**CCB380**

***Applications of MOT( Molecular Behaviour ) ; 1 Bond Order; 2 Bond Length; 3 Bond Strength; 4 Ionisation Energy; 5 Colour***

**38.** Which of the following have identical bond order ?

(A) $O_2^{2\oplus}$ (B) $NO^{\oplus}$ (C) $CN^{\ominus}$ (D) $CN^{\oplus}$

**CCB379**

***Properties Affected by Polarisation; Melting Point of Ionic Compounds***

**39.** Choose the correct order for the given properties.

(A) $NaF < MgF_2 < AlF_3$ : covalent character order.

(B) $NaF < MgF_2 < AlF_3$ : melting point order

(C) $NaF < MgF_2 < AlF_3$ : lattice energy order

(D) $NaF > MgF_2 > AlF_3$ : order of polarising power of cation.

**CCB416**

***Properties Affected by Polarisation; Solubility of Ionic Compounds in Water***

**40.** Choose the correct order(s) for the given properties.

(A) $MgSO_4 < SrSO_4 < BaSO_4$ : Thermal stability order

(B) $BeC_2O_4 < CaC_2O_4 < BaC_2O_4$ : Solubility in water

(C) $LiCl > NaCl > KCl$ : Melting point order

(D) $BeF_2 > CaF_2 > SrF_2$ : Covalent character order

**CCB414**



For More Join: @IITJEE_Advanced

## ASSERTION & REASON TYPE QUESTIONS

### *Polarity of Species*

**41.** Statement-1 : Dipole moment of $H_2O$ is more than that of $OF_2$.
Statement-2 : In $H_2O$, the resultant bond dipole moment of O – H bond and the resultant lone pair dipole moment are in opposite direction.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

**CCB303**

### *Types of Hydrogen Bonding*

**42.** Statement-1 : p-Hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid.
Statement-2 : o-Hydroxybenzoic acid has intramolecular hydrogen bonding.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement-2 is False.
(D) Statement-1 is False, Statement-2 is True.

**CCB302**

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### *Linear Combination of Atomic Orbitals*

**43.** Statement-1 : The melting point of noble gases increases as its atomic mass increases.
Statement-2 : Instantaneous dipole induced dipole attraction increases with increase in atomic mass of noble gases.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

**CCB338**

**44.** Statement-1 : $CCl_4$ has lower boiling point than that of $SiCl_4$.
Statement-2 : The magnitude of negative charge developed at chlorine atoms in $SiCl_4$ is more in comparison to negative charge developed at chlorine atoms in $CCl_4$.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

**CCB339**



For More Join: @IITJEE_Advanced

***Bonding in some homonuclear diatomic molecules (Molecular Orbital Diagram of $H_2$, $He_2$, $Li_2$, $O_2$ ,$B_2$,$C_2$,$N_2$,$F_2$ , $Ne_2$ )***

**45.** Statement-1 : Super oxide ion is paramagnetic whereas peroxide ion is diamagnetic.
Statement-2 : Super oxide ion has one unpaired electron whereas peroxide ion has no unpaired electron.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

**CCB346**

**46.** Statement-1: $H_2$ molecule is more stable than He–H molecule.
Statement-2: The antibonding electron in He–H molecule decreases the bond order and there by the stability.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

**CCB347**

***Concept of Hydration; Hydration Energy; Hydrated Radius; Ionic Mobility in Water; Conductivity in Water; Condition for Solubility of Ionic Compounds in Water***

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**47.** **Statement-1 :** Among alkali metal cations, $Li^{\oplus}$(aq.) has highest electrical conductance.
**Statement-2 :** $Li^{\oplus}$(aq.) is largest alkali metal cation because of greater degree of hydration.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

**CCB394**

## COMPREHENSION TYPE QUESTIONS

***Polarity of Species***

**Paragraph for Questions: 48 & 49**

Bond angle and bond polarity are important parameters associated for bonding in molecule.

**48.** Which of the following bond angle order is incorrect.
(A) $NH_3 > PH_3$ (B) $BF_3 < BH_3$
(C) $O(CH_3)_2 < O(C_2H_5)_2$ (D) $CO_3^{2\ominus} < CO_2$

**CCB292**

**49.** Which of the following molecule is polar and non-planar
(A) $CH_2Cl_2$ (B) $PCl_5$
(C) $SO_2$ (D) $XeF_4$

**CCB293**



For More Join: @IITJEE_Advanced

*Applications of MOT(Molecular Behaviour) ; 1 Bond Order; 2 Bond Length; 3 Bond Strength; 4 Ionisation Energy; 5 Colour*

**Paragraph for Questions 50 & 51**

Molecular orbital theory is based on linear combination of atomic orbitals (LCAO). According to LCAO when respective atomic orbitals of the atoms interact, they undergoes constructive and destructive interference giving two types of molecular orbital i.e. bonding and antibonding molecular orbitals respectively.

**50.** Which of the following specie is paramagnetic ?

(A) $NO^{\ominus}$ (B) $O_2^{2\ominus}$ (C) $CN^{\ominus}$ (D) CO

**CCB382**

**51.** Bond order of $Be_2$ is:

(A) 1 (B) 2 (C) 3 (D) 0

**CCB383**

**MATRIX MATCH TYPE QUESTION**

*VSEPR Table*

**52.** Select the correct matching:

| **List I** | **List II** |
|---|---|
| A : $XeF_4$ | 1. Pyramidal |
| B : $XeF_6$ | 2. T-shape |
| C : $XeO_3$ | 3. Distorted octahedral |
| D : $XeOF_2$ | 4. Square planar |

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| | A | B | C | D |
|---|---|---|---|---|
| (A) | 4 | 3 | 1 | 2 |
| (B) | 1 | 2 | 3 | 4 |
| (C) | 2 | 1 | 3 | 4 |
| (D) | 4 | 1 | 3 | 2 |

**CCB213**

*Bond Parameters; Bond Angle*

**53.** Match list I with list II and select the correct answer:

| **List I (species)** | **List II (O-N-O angle)** |
|---|---|
| (P) $NO_2^{\oplus}$ | (1) 180° |
| (Q) $NO_2$ | (2) 134° |
| (R) $NO_2^{\ominus}$ | (3) 120° |
| (S) $NO_3^{\ominus}$ | (4) 115° |
| | (5) 109° |

| | **P** | **Q** | **R** | **S** |
|---|---|---|---|---|
| (A) | 5 | 4 | 3 | 2 |
| (B) | 5 | 2 | 4 | 3 |
| (C) | 1 | 2 | 4 | 3 |
| (D) | 1 | 4 | 3 | 2 |

**CCB253**



For More Join: @IITJEE_Advanced

### *Odd Electron Molecules*

**54.** Match list I with list II and select the correct answer :

| **List–I (Molecule / Species)** | **List–II (Unpaired electron resides in)** |
|---|---|
| (P) $NO_2$ | (1) d-orbital |
| (Q) $ClO_2$ | (2) $sp^2$ - orbital |
| (R) $ClO_3$ | (3) $sp^3$ - orbital |
| (S) $CH_3$ | (4) p-orbital |

Code :

| | P | Q | R | S |
|---|---|---|---|---|
| (A) | 2 | 4 | 1 | 3 |
| (B) | 2 | 1 | 3 | 4 |
| (C) | 1 | 4 | 2 | 3 |
| (D) | 3 | 1 | 2 | 4 |

**CCB243**

### *Polarity of Species*

**55.** Match the column

| **Column I** | **Column II** |
|---|---|
| (A) $I(CN)_2^{\ominus}$ | (P) Having $p\pi-p\pi$ bond & $\mu_D = 0$ |
| (B) $CO_3^{2\ominus}$ | (Q) Having $p\pi-d\pi$ bond & $\mu_D \neq 0$ |
| (C) $XeO_2F_2$ | (R) Planar |
| (D) $SOF_4$ | (S) Central atom is $sp^3d$ |

(A) (A) Q,S (B) P,R (C) Q,S (D) P,R,S (B) (A) P,R,S (B) Q,S (C) P,R (D) Q,S
(C) (A) P,R,S (B) P,R (C) Q,S (D) Q,S (D) (A) Q,S (B) P,R (C) P,R,S (D) Q,S

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**CCB281**

**56.**

| **Column-I** | **Column-II** |
|---|---|
| (1) 2 lone pair on central atom | (P) $XeF_5^{\ominus}$ |
| (2) Zero dipole moment | (Q) $NF_3$ |
| (3) Planar | (R) $C_2H_4$ |
| (4) All adjacent bond angles are equal | (S) $XeF_4$ |

(A) (1) P, Q; (2) P, R, S; (3) P, R, S; (4) P, Q, R (B) (1) P, S; (2) P, R, S; (3) P, R, S; (4) P, Q, S
(C) (1) R, S; (2) P, Q, S; (3) P, R, S; (4) P, Q, S (D) (1) R, S; (2) P, R, S; (3) P, R, S; (4) P, Q, R

**CCB304**

### *Bridge Bonding(3C-2e- , 3C-4e-) , $B_2H_6$ , $Al_2Cl_6$ , Bond Behaviour of $B_2H_6$ ; Bridge Bonding in other Compounds; Reactions of Diborane*

**57.** Match the column :

| **Column I** | **Column II** |
|---|---|
| (A) $B_2H_6$ | (P) (3c-4e$^{\ominus}$) bond |
| (B) $Be_2H_4$ | (Q) (3c-2e$^{\ominus}$) bond |
| (C) $Be_2Cl_4$ | (R) Central atom is $sp^3$ hybridised |
| (D) $Al_2(CH_3)_6$ | (S) central atom is $sp^2$ hybridised. |

(A) A $\rightarrow$ P, R; B $\rightarrow$ Q, S; C $\rightarrow$ P, S; D $\rightarrow$ Q,R (B) A $\rightarrow$ Q, R; B $\rightarrow$ P, S; C $\rightarrow$ P, Q; D $\rightarrow$ P,R
(C) A $\rightarrow$ Q, R; B $\rightarrow$ Q, S; C $\rightarrow$ P, S; D $\rightarrow$ P,R (D) A $\rightarrow$ S, R; B $\rightarrow$ R, S; C $\rightarrow$ P, S; D $\rightarrow$ P,R

**CCB536**



For More Join: @IITJEE_Advanced

*Introduction of Weak Forces; Types of Weak Forces; Factors Affecting London Dispersion Forces; Comparison of Boiling Point of Species*

**58.** Match list I with list II and select the correct answer:

| **List–I (Process)** | **List–II (Operating intraction involved)** |
|---|---|
| (P) Clathrate compound of Xe in ice | (1) Ion - Dipole |
| (Q) Liquation of Xe gas | (2) Dipole - Dipole |
| (R) Liquation of HCl gas | (3) Dipole - Induced dipole |
| (S) Hydration of $Na^{\oplus}$ | (4) London forces |

**Code :**

| | **P** | **Q** | **R** | **S** |
|---|---|---|---|---|
| (A) | 1 | 3 | 4 | 2 |
| (B) | 3 | 4 | 1 | 2 |
| (C) | 3 | 4 | 2 | 1 |
| (D) | 1 | 3 | 2 | 4 |

**CCB328**

**TABLE TYPE QUESTION**

*Linear Combination of Atomic Orbitals*

**Direction for Q.59 to Q.60 by appropriately matching the information given in the three columns of the following table**

| Molecular Orbital | Number of nodal plane | Symmetry of Molecular Orbital |
|---|---|---|
| (1) $\sigma_s$ | (P) 3 | (I) BMO, gerade |
| (2) $\pi_p$ | (Q) 2 | (II) ABMO, ungerade |
| (3) $\sigma^*_{pZ}$ | (R) 1 | (III) BMO, ungerade |
| (4) $\delta^*$ | (S) 0 | (IV) ABMO, gerade |

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**59.** Which of the following matching is **INCORRECT.**

(A) (1), S, I (B) (2), R, I (C) (3), R, II (D) (4), P, II

**CCB384**

**60.** How many orbitals are occupied with set (1)(S)(I) in $F_2$

(A) 0 (B) 1 (C) 2 (D) 3

**CCB385**



For More Join: @IITJEE_Advanced

# EXERCISE (S)

## Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

### *Kossel -Lewis approach to Chemical Bonding; Lewis Dot Structure*

**1.** Draw the Lewis dot structure of the given compounds and find how many compounds contain lone pair(s) on central atom ?

| | | | |
|---|---|---|---|
| 1. $CCl_3^{\ominus}$ | 2. $CCl_4$ | 3. $NO_2^{\ominus}$ | 4. $NO_3^{\ominus}$ |
| 5. CO | 6. $CO_2$ | 7. $SCN^{\ominus}$ | 8. $SnCl_3^{\ominus}$ |
| 9. $BH_4^{\ominus}$ | 10. $BeF_4^{2\ominus}$ | 11. $SO_3$ | 12. $H_3O^{\oplus}$ |
| 13. $CO_3^{2\ominus}$ | | | |

**CCB157**

### *Limitations of Lewis Dot Structure (Hypervalent ,Hypovalent , Odd Electron Species)*

**2.** How many of the following compounds are exception to the octet rule ?
$H_2SO_4$, $BeCl_2$, $ClO_2$, $H_2O$, $H_2$, $CCl_4$, HF, $NH_3$, $BF_3$

**CCB156**

### *Types of Covalent Bond and Overlapping of Orbitals; Orbital Overlap along Different Axes*

**3.** Which of the given set of orbitals form 'σ' bond between two atoms along y-axis ?
$s + p_x$, $s + p_y$, $p_y + p_y$, $p_x + p_y$, $p_y + p_z$, $p_x + p_z$

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**CCB181**

### *Prediction of Hybridisation; Types of Hybridisation*

**4.** In which of the given species central atom is $sp^3$ hybridized ?
$SF_4$, $POCl_3$, $H_2O$, $XeO_3$, $PCl_5$, $OF_2$, $SO_3^{2\ominus}$, $XeO_3F_2$, $XeF_4$

**CCB203**

### *Calculation of Number of p pi-p pi and p pi-d pi Bonds*

**5.** Find the number of $p\pi - p\pi$ bond in $IO_3^{\ominus}$ :

**CCB204**

**6.** Find out total number of $p\pi - d\pi$ bonds present in $SO_2$ molecule :

**CCB205**

### *Solid State Hybridisation*

**7.** What is the bond angle difference between anionic part of solid Iodine molecule and anionic part of $N_2O_5(s)$

**CCB256**

### *VSEPR Table*

**8.** How many compounds have distorted shape ?
$CH_4$, $CHCl_3$, $CH_2Cl_2$, $SF_6$, $BF_2Cl$

**CCB228**



For More Join: @IITJEE_Advanced

**9.** Among the following total number of planar molecules is / are :

$Cl_2O$, $P(CH_3)_3$, $N(CH_3)_3$, $ClO_2^{\oplus}$, $\overset{\oplus}{C}H_3$, $NCl_3$

**CCB234**

### *Maximum Atoms in A Plane; Number of Such Planes*

**10.** Among the given species, how many species have maximum '3' atoms lying in a plane?
$BeCl_2$, $SnCl_2$, $SF_2$, $XeO_3F_2$, $IF_7$

**CCB229**

**11.** Find the maximum number of atoms lying in a plane in $AsF_5$ :

**CCB231**

### *Bond Parameters; Bond length*

**12.** Find the total number of following molecule(s) which have all bond lengths are same :
$XeF_4$, $SF_4$, $SH_2$, $NO_3^{\ominus}$, $SiF_4$, $ClF_3$, $PF_2Cl_3$, $XeO_3F_2$

**CCB232**

### *Odd Electron Molecules*

**13.** Find the number of species which have tendency to dimerize :
$\dot{N}O_2$, $\dot{C}lO_3$, $\dot{C}lO_2$, $\dot{C}H_3$, $\dot{C}F_3$

**CCB257**

### *Dipole Moment; Calculation of Percent Ionic Character*

**14.** A diatomic molecule has a dipole moment of 1.2 D. If bond length is 1.0 Å, what percentage of an electronic charge exists on each atom.
[Add the digits untill you are getting single digit. e.g. If the answer is 537, then 5 + 3 + 7 = 15] again 1 + 5 = 6 and 6 is to be written as answer.

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**CCB294**

### *Polarity of Species*

**15.** Among the given molecules, number of molecules having some net polarity ($\mu \neq 0$)
$XeF_4$, $SO_2$, $SF_4$, $CH_2Cl_2$, $NF_3$, $BF_3$

**CCB417**

### *Types of Hydrogen Bonding*

**16.** Number of species from the following which shows hydrogen bonding.
Chloral, HF, $H_2O_2$, $HPO_3^{2\ominus}$, $HNO_3$, $H_3BO_3$, $NaHCO_3$, $HF_2^{\ominus}$

**CCB318**

### *Back Bonding*

**17.** In how many of the following species back bonding is possible ?
$PH_3$, $BF_3$, $BCl_3$, $PF_3$, $BF_4^{\ominus}$, $N(SH_3)_3$, $N(CH_3)_3$

**CCB537**

### *Bridge Bonding(3C-2e- , 3C-4e-) , $B_2H_6$ , $Al_2Cl_6$ , Bond Behaviour of B2H6 ; Bridge Bonding in other Compounds; Reactions of Diborane*

**18.** Number of lone pair present in $Al_2Cl_6$.

**CCB538**

**19.** Total number of Non-ionisable Hydrogen atom in $H_3PO_3$.

**CCB539**

**20.** Basicity of $H_4P_2O_7$.

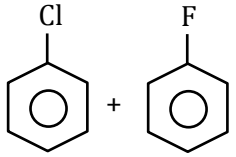
**CCB540**



For More Join: @IITJEE_Advanced

***Introduction of Weak Forces; Types of Weak Forces; Factors Affecting London Dispersion Forces; Comparison of Boiling Point of Species***

**21.** Dipole-Dipole interaction is present in how many of the given species ?

(i) $H_2O + H_2O$ (ii) $CCl_4 + C_6H_6$ (iii) $CHCl_3 + CHCl_3$ (iv) 


(v) $K^+ + I^-$ (vi) He + Ne

**CCB419**

***Energy level diagram for s-p mixing and without s-p mixing case; Electronic Configuration***

**22.** If s and p-orbital mixing is operative for all species given, then how many following species is/are paramagnetic in nature?

$O_2^-$, $N_2$, $C_2$, $CN^-$, $B_2$, $O_2^{2+}$

**CCB420**

***Bonding in some homonuclear diatomic molecules (Molecular Orbital Diagram of $H_2$, $He_2$, $Li_2$, $O_2$ ,$B_2$,$C_2$,$N_2$,$F_2$ , $Ne_2$)***

**23.** Total number of species in which highest occupied molecular orbital has gerade type symmetry.

$B_2$ ,$N_2$ ,$C_2$ ,$N_2^{2+}$ ,$O_2$ ,$O_2^{2-}$ ,$N_2^{2-}$

**CCB421**

**24.** How many of the following molecule(s) or ion(s) is/are having bond order less than or equal to two?

$B_2$, $C_2$, $O_2^{2-}$ , $O_2^+$ , $H_2^+$ , $N_2^+$ , $O_2^-$ , $H_2$

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**CCB422**

***Applications of MOT( Molecular Behaviour ) ; 1 Bond Order; 2 Bond Length; 3 Bond Strength; 4 Ionisation Energy; 5 Colour***

**25.** Find the number of Chemical species which are paramagnetic in nature.

$O_2^{2-}$ , NO, $ClO_2$, OF, $B_2$

**CCB424**

## Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

***Prediction of Hybridisation; Types of Hybridisation***

**26.** Find our number of transformation among following which involves the change of hybridisation of underlines atom.

(A) $\underline{H_2O} + H^+ \rightarrow \underline{H_3O}^+$ (B) $NH_3 + \underline{B}F_3 \rightarrow NH_3 \cdot \underline{B}F_3$

(C) $\underline{Xe}F_6 \rightarrow \underline{Xe}F_5^+ + F^-$ (D) $2\underline{P}Cl_5 \rightarrow \underline{P}Cl_4^+ + PCl_6^-$

(E) $\underline{C}H_3 - CH_3 \rightarrow \underline{C}H_3^- + CH_3^+$

**CCB505**



For More Join: @IITJEE_Advanced

### *Maximum Atoms in A Plane; Number of Such Planes*

**27.** Find the maximum number of plane having maximum number of atoms in $CCl_4$.

**CCB506**

### *Concept of Resonance; Bond Order; Comparison of Bond Length and Bond Strength using Bond Order*

**28.** If x = C–O bond order in $CO_3^{2-}$

y = C–O bond order in $HCOO^-$

z = Cl–O bond order in $ClO_4^-$

Then x + y + z = ....

**CCB507**

### *Dipole Moment; Calculation of Percent Ionic Character*

**29.** Experimental dipole moment of HCl molecule is 1.03 D and bond distance is 1.275 Å. Calculate the value of x/10. Where x is % of ionic character :

**CCB508**

**30.** Calculate the percentage of covalent character of HX having bond length = 1.62 Å and observed dipole moment = 0.39 D.

**CCB509**

### *Back Bonding*

**31.** Find the number molecules in which bond angle is changed due to back bonding w.r.t B/O/N.

$H_3BO_3$, $B(OMe)_3$ , $BF_3$, $N(SiH_3)_3$ , $O(SiH_3)_2$

**CCB510**

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### *Oxyacids of Phosphorus*

**32.** Find the value of z in $H_4P_2O_8$.

$$z = \frac{x+d}{a+c+b}$$

Total number of σ bond = x
Total number of lone pair of e⁻s are = a
Total number of $sp^3$ hybridised atoms are b
Total number of pπ-pπ bonds are c
Total number of pπ-dπ bonds are d

**CCB511**

### *Oxyacids of Sulphur*

**33.** Find the value of $\frac{\text{sigma bond} + \text{sum of total lone pair } e^-}{\text{Total number of } \pi \text{ bond}}$ for Marshall's acid.

**CCB512**

### *Applications of MOT( Molecular Behaviour ) ; 1 Bond Order; 2 Bond Length; 3 Bond Strength; 4 Ionisation Energy; 5 Colour*

**34.** There are some species given below.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| (i) | $O_2^+$ | (ii) | CO | (iii) | $B_2$ | (iv) | $O_2^-$ | (v) | $NO^+$ |
| (vi) | $He_2^+$ | (vii) | $C_2^{2+}$ | (viii) | $CN^-$ | (ix) | $N_2^-$ | | |

Total number of species which have their fractional bond order

**CCB513**

**35.** Total number of P—H bonds in cyclotrimetaphosphoric acid.

**CCB541**

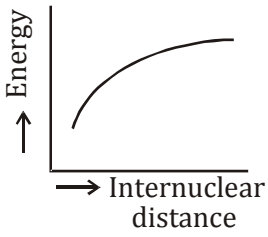


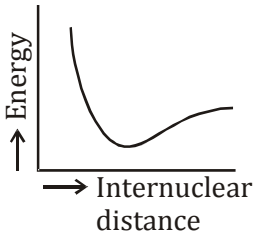
For More Join: @IITJEE_Advanced

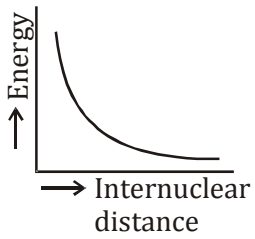
# EXERCISE - JEE (Main) PYQ

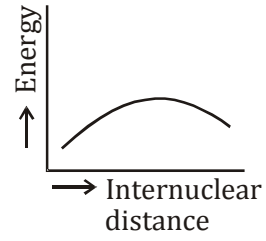
**1.** The potential energy curve for the $H_2$ molecule as a function of internuclear distance is :

**[JEE (Main) 2020]**

(1) 


(2) 

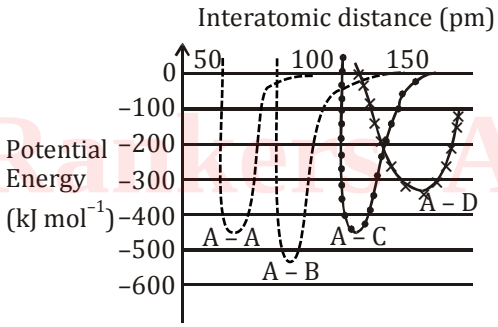

(3) 


(4) 


**CCB039**

**2.** The intermolecular potential energy for the molecules A, B, C and D given below suggests that :

**[JEE (Main) 2020]**




(1) D is more electronegative than other atoms

(2) A-D has the shortest bond length

(3) A-B has the stiffest bond

(4) A–A has the largest bond enthalpy

**CCB043**

**3.** Match the **LIST-I** with **LIST-II**

| **LIST-I (Classification of molecules based on octet rule)** | | **LIST-II (Example)** | |
|---|---|---|---|
| A. | Molecules obeying octet rule | I. | NO, $NO_2$ |
| B. | Molecules with incomplete octet | II. | $BCl_3$, $AlCl_3$ |
| C. | Molecules with incomplete octet with odd electron | III. | $H_2SO_4$, $PCl_5$ |
| D. | Molecules with expanded octet | IV. | $CCl_4$, $CO_2$ |

Choose the **correct** answer from the options given below : **[JEE (Main) 2025]**

(1) A-IV, B-II, C-I, D-III

(2) A-III, B-II, C-I, D-IV

(3) A-IV, B-I, C-III, D-II

(4) A-II, B-IV, C-III, D-I

**CCB598**



For More Join: @IITJEE_Advanced

**4.** Number of compounds with one lone pair of electrons on central atom amongst following is _
$O_3$, $H_2O$, $SF_4$, $ClF_3$, $NH_3$, $BrF_5$, $XeF_4$

[JEE (Main) 2024]

**CCB514**

**5.** The hybridisations of the atomic orbitals of nitrogen in $NO_2^{\ominus}$, $NO_2^{\oplus}$ and $NH_4^{\oplus}$ respectively are.

[JEE (Main) 2021]

(1) $sp^3$, $sp^2$ and sp (2) sp, $sp^2$ and $sp^3$ (3) $sp^3$, sp and $sp^2$ (4) $sp^2$, sp and $sp^3$

**CCB047**

**6.** The structure of $PCl_5$ in the solid state is : [JEE (Main) 2020]

(1) Square pyramidal
(2) Tetrahedral $[PCl_4]^{\oplus}$ and octahedral $[PCl_6]^{\ominus}$
(3) Square planar $[PCl_4]^{\oplus}$ and octahedral $[PCl_6]^{\ominus}$
(4) Trigonal bipyramidal

**CCB038**

**7.** The number of molecules/ions that show linear geometry among the following is _____ .
$SO_2$, $BeCl_2$, $CO_2$, $N_3^-$, $NO_2$, $F_2O$, $XeF_2$, $NO_2^+$, $I_3^-$, $O_3$ [JEE (Main) 2025]

**CCB599**

**8.** The molecules having square pyramidal geometry are [JEE (Main) 2025]

(1) $BrF_5$ & $XeOF_4$ (2) $SbF_5$ & $XeOF_4$
(3) $SbF_5$ & $PCl_5$ (4) $BrF_5$ & $PCl_5$

**CCB600**

**9.** Consider 'n' is the number of lone pair of electrons present in the equatorial position of the most stable structure of $ClF_3$. The ions from the following with 'n' number of unpaired electrons are :

A. $V^{3+}$ B. $Ti^{3+}$
C. $Cu^{2+}$ D. $Ni^{2+}$
E. $Ti^{2+}$

Choose the ***correct*** answer from the options given below : [JEE (Main) 2025]

(1) A and C only (2) A, D and E only
(3) B and C only (4) B and D only

**CCB601**

**10.** Which of the following statement is true with respect to $H_2O$, $NH_3$ and $CH_4$?

A. The central atoms of all the molecules are $sp^3$ hybridized.
B. The H–O–H, H–N–H and H–C–H angles in the above molecules are 104.5º, 107.5º and 109.5º respectively.
C. The increasing order of dipole moment is $CH_4 < NH_3 < H_2O$.
D. Both $H_2O$ and $NH_3$ are Lewis acids and $CH_4$ is a Lewis base
E. A solution of $NH_3$ in $H_2O$ is basic. In this solution $NH_3$ and $H_2O$ act as Lowry-Bronsted acid and base respectively.

Choose the correct answer from the options given below : [JEE (Main) 2025]

(1) A, B and C only (2) C, D and E only
(3) A, D and E only (4) A, B, C and E only

**CCB602**



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**11.** Match **List I** and with **List II**

| List-I (Molecule) | | List-II (Shape) | |
|---|---|---|---|
| A | $NH_3$ | I | Square pyramid |
| B | $BrF_5$ | II | Tetrahedral |
| C | $PCl_5$ | III | Trigonal pyramidal |
| D | $CH_4$ | IV | Trigonal bipyramidal |

Choose the **correct** answer from the option below: **[JEE (Main) 2024]**

(1) A-IV, B-III, C-I, D-II (2) A-II, B-IV, C-I, D-III (3) A-III, B-I, C-IV, D-II (4) A-III, B-IV, C-I, D-II

**CCB515**

**12.** Match **List I** with **List II**

| LIST–I (molecules/ions) | | LIST–II (No. of lone pairs of e⁻ on central atom) | |
|---|---|---|---|
| (A) | $IF_7$ | I | Three |
| (B) | $ICl_4^-$ | II | One |
| (C) | $XeF_6$ | III | Two |
| (D) | $XeF_2$ | IV | Zero |

Choose the **correct** answer from the options given below: **[JEE (Main) 2023]**

(1) A – II, B – III, C – IV, D – I (2) A – IV, B – III, C – II, D – I

(3) A – II, B – I, C – IV, D – III (4) A – IV, B – I, C – II, D – III

**CCB426**

**13.** Amongst the following, the number of species having the linear shape is _____.

$XeF_2, I_3^+, C_3O_2, I_3^-, CO_2, SO_2$, BeCl and $BCl_2^{\ominus}$ **[JEE (Main) 2023]**

**CCB427**

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**14.** Match List-I with List-II :

| List-I Species | List-II Geometry/Shape |
|---|---|
| A. $H_3O^+$ | I. Tetrahedral |
| B. Acetylide anion | II. Linear |
| C. $NH_4^+$ | III. Pyramidal |
| D. $ClO_2^-$ | IV. Bent |

Choose the correct answer from the options given below : **[JEE (Main) 2023]**

(1) A-III, B-II, C-I, D-IV (2) A-III, B-I, C-II, D-IV (3) A-III, B-IV, C-I, D-II (4) A-III, B-IV, C-II, D-I

**CCB429**

**15.** Match **List - I** with **List - II**.

| **List - I (Compound)** | **List – II (Shape)** |
|---|---|
| (A) $BrF_5$ | (I) bent |
| (B) $[CrF_6]^{3\ominus}$ | (II) square pyramidal |
| (C) $O_3$ | (III) trigonal bipyramidal |
| (D) $PCl_5$ | (IV) octahedral |

Choose the **correct** answer from the options given below : **[JEE (Main) 2022]**

(1) (A) – (I), (B) - (II), (C) - (III), (D) - (IV) (2) (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

(3) (A) - (II), (B) - (IV), (C) - (I), (D) - (III) (4) (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

**CCB048**



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**16.** The shape/structure of $[XeF_5]^{\ominus}$ and $XeO_3F_2$, respectively, are : **[JEE (Main) 2020]**

(1) Pentagonal planar and trigonal bipyramidal
(2) Trigonal bipyramidal and pentagonal planar
(3) Octahedral and square pyramidal
(4) Trigonal bipyramidal and trigonal bipyramidal

**CCB045**

**17.** The compound that has the largest H–M–H bond angle (M = N, O, S, C), is : **[JEE (Main) 2020]**

(1) $H_2O$ (2) $CH_4$ (3) $NH_3$ (4) $H_2S$

**CCB040**

**18.** Given below are two statements :

**Statement(I) :** Experimentally determined oxygen-oxygen bond lengths in the $O_3$ are found to be same and the bond length is greater than that of a O = O (double bond) but less than that of a single (O – O) bond.

**Statement (II) :** The strong lone pair-lone pair repulsion between oxygen atoms is solely responsible for the fact that the bond length in ozone is smaller than that of a double bond (O=O) but more than that of a single bond (O – O).

In the light of the above statements, choose the **correct** answer from the options given below:

**[JEE (Main) 2025]**

(1) **Statement I** is true but **Statement II** is false
(2) Both **Statement I** and **Statement II** are true
(3) Both **Statement I** and **Statement II** are false
(4) **Statement I** is false but **Statement II** is true

**CCB603**

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**19.** Arrange the following compounds in increasing order of their dipole moment :

HBr, $H_2S$, $NF_3$ and $CHCl_3$ **[JEE (Main) 2025]**

(1) $NF_3 < HBr < H_2S < CHCl_3$ (2) $HBr < H_2S < NF_3 < CHCl_3$
(3) $H_2S < HBr < NF_3 < CHCl_3$ (4) $CHCl_3 < NF_3 < HBr < H_2S$

**CCB604**

**20.** The number of non-polar molecules from the following is _____

HF, $H_2O$, $SO_2$, $H_2$, $CO_2$, $CH_4$, $NH_3$, HCl, $CHCl_3$, $BF_3$

**[JEE (Main) 2024]**

**CCB516**

**21.** If $AB_4$ molecule is a polar molecule, a possible geometry of $AB_4$ is : **[JEE (Main) 2020]**

(1) Square pyramidal (2) Tetrahedral
(3) Square planar (4) Rectangular planar

**CCB044**

**22.** If the boiling point of $H_2O$ is 373 K, the boiling point of $H_2S$ will be : **[JEE (Main) 2020]**

(1) Greater than 300 K but less than 373 K (2) Less than 300 K
(3) Equal to 373 K (4) More than 373 K

**CCB042**

**23.** Choose the correct stability order of group 13 elements in their +1 oxidation state.

**[JEE (Main) 2022]**

(1) Al < Ga < In < Tl (2) Tl < In < Ga < Al (3) Al < Ga < Tl < In (4) Al < Tl < Ga < In

**CCB542**



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**24.** The Lewis acid character of boron tri halides follows the order : **[JEE (Main) 2023]**

(1) $BBr_3 > BI_3 > BCl_3 > BF_3$ (2) $BCl_3 > BF_3 > BBr_3 > BI_3$

(3) $BF_3 > BCl_3 > BBr_3 > BI_3$ (4) $BI_3 > BBr_3 > BCl_3 > BF_3$

**CCB543**

**25.** Identify the correct statement for $B_2H_6$ from those given below. **[JEE (Main) 2022]**

(A) In $B_2H_6$, all B-H bonds are equivalent.

(B) In $B_2H_6$ there are four 3-centre-2-electron bonds.

(C) $B_2H_6$ is a Lewis acid.

(D) $B_2H_6$ can be synthesized form both $BF_3$ and $NaBH_4$.

(E) $B_2H_6$ is a planar molecule.

Choose the most appropriate answer from the options given below:

(1) (A) and (E) only (2) (B), (C) and (E) only

(3) (C) and (D) only (4) (C) and (E) only

**CCB544**

**26.** The correct statement about $B_2H_6$ is: **[JEE (Main) 2021]**

(1) Terminal B–H bonds have less p-character when compared to bridging bonds.

(2) The two B–H–B bonds are not of same length

(3) All B–H–B angles are of 120°

(4) Its fragment, $BH_3$, behaves as a Lewis base

**CCB545**

**27.** Sum of π-bonds present in peroxodisulphuric acid and pyrosulphuric acid is **[JEE (Main) 2023]**

**CCB050**

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**28.** Consider the following sulphure based oxoacids.

$H_2SO_3$, $H_2SO_4$, $H_2S_2O_8$ and $H_2S_2O_7$.

Amongst these oxoacids, the number of those with peroxo(O-O) bond is_____.

**[JEE (Main) 2022]**

**CCB546**

**29.** Match the type of interaction in Column A with the distance dependence of their interaction energy in Column B : **[JEE (Main) 2020]**

| **Column-A** | **Column-B** |
|---|---|
| (I) Ion - ion | (a) $\frac{1}{r}$ |
| (II) Dipole - dipole | (b) $\frac{1}{r^2}$ |
| (III) London dispersion | (c) $\frac{1}{r^3}$ |
| | (d) $\frac{1}{r^6}$ |

**Code:**

(1) (I)-(a), (II)-(b), (III)-(c) (2) (I)-(a), (II)-(c), (III)-(d)

(3) (I)-(a), (II)-(b), (III)-(d) (4) (I)-(b), (II)-(d), (III)-(c)

**CCB041**



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**30.** Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**
**Assertion A :** Zero orbital overlap is an out of phase overlap.
**Reason :** It results due to different orientation/ direction of approach of orbitals.
In the light of the above statements. Choose the ***correct*** answer from the options given below
**[JEE (Main) 2022]**

(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true but R is NOT the correct explanation of A
(3) A is true but R is false
(4) A is false but R is true

**CCB049**

**31.** Which of the following linear combination of atomic orbitals will lead to formation of molecular orbitals in homonuclear diatomic molecules [internuclear axis in z-direction] ?
**A.** $2p_z$ and $2p_x$
**B.** 2s and $2p_x$
**C.** $3d_{xy}$ and $3d_{x^2-y^2}$
**D.** 2s and $2p_z$
**E.** $2p_z$ and $3d_{x^2-y^2}$

**[JEE (Main) 2025]**

(1) E Only
(2) A and B Only
(3) D Only
(4) C and D Only

**CCB605**

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**32.** What is the number of unpaired electron(s) in the highest occupied molecular orbital of the following species : $N_2$: $N_2^+$; $O_2$; $O_2^+$ ? **[JEE (Main) 2023]**

(1) 0, 1, 2, 1
(2) 2, 1, 2, 1
(3) 0, 1, 0, 1
(4) 2, 1, 0, 1

**CCB425**

**33.** The number of species from the following which are paramagnetic and with bond order equal to one is___.
$H_2, He_2^+, O_2^+, N_2^{2-}, O_2^{2-}, F_2, Ne_2^+, B_2$ **[JEE (Main) 2024]**

**CCB517**

**34.** The bond order and magnetic property of acetylide ion are same as that of **[JEE (Main) 2023]**

(1) $NO^+$
(2) $O_2^+$
(3) $O_2^-$
(4) $N_2^+$

**CCB430**

**35.** In which of the following processes, the bond order increases and paramagnetic character changes to diamagnetic one? **[JEE (Main) 2023]**

(1) $O_2 \rightarrow O_2^{2-}$
(2) $NO \rightarrow NO^+$
(3) $N_2 \rightarrow N_2^+$
(4) $O_2 \rightarrow O_2^+$

**CCB431**

**36.** The number of following factors which affect the percent covalent character of the ionic bond is___
(a) Polarising power of cation
(b) Extent of distortion of anion
(c) Polarisability of the anion
(d) Polarising power of anion **[JEE (Main) 2023]**

**CCB428**



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# EXERCISE - JEE (Advanced) PYQ

**1.** Which ordering of compounds is according to the decreasing order of the oxidation state of Nitrogen ? **[JEE (Advanced) 2012]**

(A) $HNO_3$, NO, $NH_4Cl$, $N_2$ (B) $HNO_3$, NO, $N_2$, $NH_4Cl$

(C) $HNO_3$, $NH_4Cl$, NO, $N_2$ (D) NO, $HNO_3$, $NH_4Cl$, $N_2$

**CCB075**

**2.** The option(s) in which at least three molecules follow Octet Rule is(are) **[JEE (Advanced) 2024]**

(A) $CO_2$, $C_2H_4$, NO and HCl (B) $NO_2$, $O_3$, HCl and $H_2SO_4$

(C) $BCl_3$, NO, $NO_2$ and $H_2SO_4$ (D) $CO_2$, $BCl_3$, $O_3$ and $C_2H_4$

**CCB518**

**3.** The total number of lone pairs of electrons in $N_2O_3$ is : **[JEE (Advanced) 2015]**

**CCB081**

**4.** The total number of lone-pairs of electrons in Melamine is : **[JEE (Advanced) 2013]**

**CCB077**

**5.** In allene ($C_3H_4$), the type(s) of hybridisation of the carbon atoms is (are) : **[JEE (Advanced) 2012]**

(A) $sp$ and $sp^3$ (B) $sp$ and $sp^2$

(C) Only $sp^2$ (D) $sp^2$ and $sp^3$

**CCB074**

**6.** Based on VSEPR model, match the xenon compounds given in List-I with the corresponding geometries and the number of lone pairs on xenon given in List-II and choose the correct option. **[JEE (Advanced) 2024]**

| | List-I | | List-II |
|---|---|---|---|
| (P) | $XeF_2$ | (1) | Trigonal bipyramidal and two lone pair of electrons |
| (Q) | $XeF_4$ | (2) | Tetrahedral and one lone pair of electrons |
| (R) | $XeO_3$ | (3) | Octahedral and two lone pair of electrons |
| (S) | $XeO_3F_2$ | (4) | Trigonal bipyramidal and no lone pair of electrons |
| | | (5) | Trigonal bipyramidal and three lone pair of electrons |

(A) P $\rightarrow$ 5, Q $\rightarrow$ 2, R $\rightarrow$ 3, S $\rightarrow$ 1 (B) P $\rightarrow$ 5, Q $\rightarrow$ 3, R $\rightarrow$ 2, S $\rightarrow$ 4

(C) P $\rightarrow$ 4, Q $\rightarrow$ 3, R $\rightarrow$ 2, S $\rightarrow$ 1 (D) P $\rightarrow$ 4, Q $\rightarrow$ 2, R $\rightarrow$ 5, S $\rightarrow$ 3

**CCB519**

**7.** The sum of the number of lone pairs of electrons on each central atom in the following species is :
$[TeBr_6]^{2\ominus}$, $[BrF_2]^{\oplus}$, $SNF_3$ and $[XeF_3]^{\ominus}$
[Atomic number : N = 7, F = 9, S = 16, Br = 35, Te = 52, Xe = 54]

**[JEE (Advanced) 2017]**

**CCB086**

**8.** The compound(s) with TWO lone pairs of electrons on the central atom is(are) : **[JEE (Advanced) 2016]**

(A) $BrF_5$ (B) $ClF_3$

(C) $XeF_4$ (D) $SF_4$

**CCB083**



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**9.** Among the triatomic molecules/ions, $BeCl_2$, $N_3^{\ominus}$, $N_2O$, $NO_2^{\oplus}$, $O_3$, $SCl_2$, $ICl_2^{\ominus}$, $I_3^{\ominus}$ and $XeF_2$ the total number of linear molecules(s) / ion(s) where the hybridization of the central atoms does not have contribution from the d-orbital(s) is :
(Atomic number : S = 16, Cl = 17, I = 53 and Xe = 54) **[JEE (Advanced) 2015]**

**CCB080**

**10.** The shape of $XeO_2F_2$ molecule is : **[JEE (Advanced) 2012]**

(A) Trigonal bipyramidal (B) Square planar
(C) Tetrahedral (D) See-saw

**CCB076**

**11.** Based on VSEPR theory, the number of 90 degree F–Br–F angles in $BrF_5$ is :

**[JEE (Advanced) 2010]**

**CCB072**

**12.** Consider the following compounds in the liquid form :
$O_2$, HF, $H_2O$, $NH_3$, $H_2O_2$, $CCl_4$, $CHCl_3$, $C_6H_6$, $C_6H_5Cl$
When a charged comb is brought near their flowing stream, how many of them show deflection as per the following figure ?



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**[JEE (Advanced) 2020]**

**CCB089**

**13.** Each of the following options contains a set of four molecules. Identify the option(s) where all four molecules possess permanent dipole moment at room temperature :

(A) $BeCl_2$, $CO_2$, $BCl_3$, $CHCl_3$ (B) $SO_2$, $C_6H_5Cl$, $H_2Se$, $BrF_5$
(C) $BF_3$, $O_3$, $SF_6$, $XeF_6$ (D) $NO_2$, $NH_3$, $POCl_3$, $CH_3Cl$

**[JEE (Advanced) 2019]**

**CCB087**

**14.** Among the following, the correct statement(s) is/are :- **[JEE (Advanced) 2017]**

(A) $Al(CH_3)_3$ has the three-centre two-electron bonds in its dimeric structure
(B) $AlCl_3$ has the three-centre two-electron bonds in its dimeric structure
(C) $BH_3$ has the three-centre two-electron bonds in its dimeric structure
(D) The Lewis acidity of $BCl_3$ is greater than that of $AlCl_3$

**CCB547**

**15.** The compound (s) having peroxide linkage is(are) :

(A) $H_2S_2O_7$ (B) $H_2S_2O_8$
(C) $H_2S_2O_5$ (D) $H_2SO_5$

**[JEE (Advanced) 2024]**

**CCB520**



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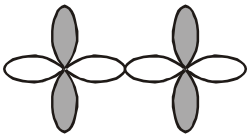
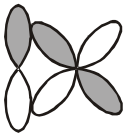
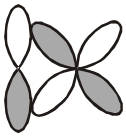
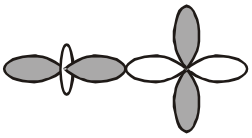
**16.** Consider the following molecules : $Br_3O_8$, $F_2O$, $H_2S_4O_6$, $H_2S_5O_6$, and $C_3O_2$.
Count the number of atoms existing in their zero oxidation state in each molecule. Their sum is___.
**[JEE (Advanced) 2023]**
**CCB548**

**17.** Among $B_2H_6$, $B_3N_3H_6$, $N_2O$, $N_2O_4$, $H_2S_2O_3$ and $H_2S_2O_8$, the total number of molecules containing covalent bond between two atoms of the same kind is :
**[JEE (Advanced) 2019]**
**CCB088**

**18.** The correct statement(s) regarding, (i) HClO, (ii) $HClO_2$, (iii) $HClO_3$ and (iv) $HClO_4$, is(are) :
**[JEE (Advanced) 2015]**
(A) The number of Cl = O bonds in (ii) and (iii) together is two
(B) The number of lone pairs of electrons on Cl in (ii) and (iii) together is three
(C) The hybridization of Cl in (iv) is $sp^3$
(D) Amongst (i) to (iv), the strongest acid is (i)
**CCB082**

**19.** For diatomic molecules, the correct statement(s) about the molecular orbitals formed by the overlap of two $2p_z$ orbitals is(are) **[JEE (Advanced) 2022]**
(A) σ orbital has a total of two nodal planes.
(B) σ* orbital has one node in the xz-plane containing the molecular axis.
(C) π orbital has one node in the plane which is perpendicular to the molecular axis and goes through the center of the molecule.
(D) π* orbital has one node in the *xy*-plane containing the molecular axis.
**CCB090**

**20.** Match the orbital overlap figures shown in List-I with the description given in List-II and select the correct answer using the code given below the lists : **[JEE (Advanced) 2014]**

| List-I | List-II |
|---|---|
| (P)  | (1) p – d π antibonding |
| (Q)  | (2) d – d σ bonding |
| (R)  | (3) p – d π bonding |
| (S)  | (4) d – d σ antibonding |

(A) P-2, Q-1, R-3, S-4
(B) P-4, Q-3, R-1, S-2
(C) P-2, Q-3, R-1, S-4
(D) P-4, Q-1, R-3, S-2
**CCB079**



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**21.** Assuming 2s-2p mixing is NOT operative, the paramagnetic species among the following is :

[JEE (Advanced) 2014]

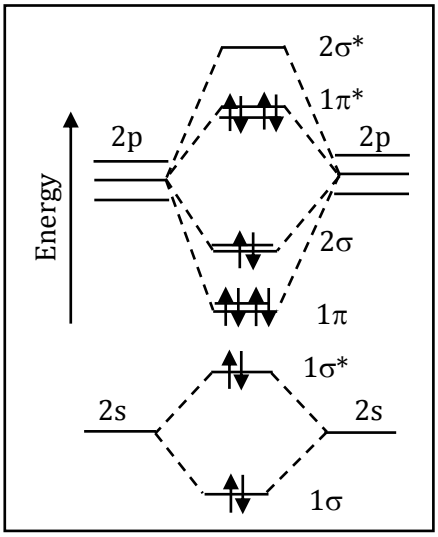
(A) $Be_2$ (B) $B_2$ (C) $C_2$ (D) $N_2$

**CCB078**

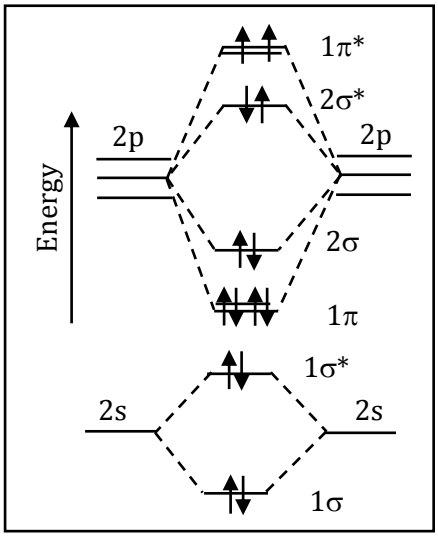
**22.** The correct molecular orbital diagram for $F_2$ molecule in the ground state is

[JEE (Advanced) 2023]

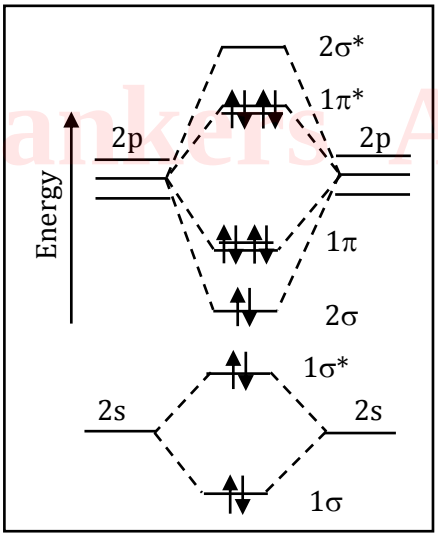
(A)



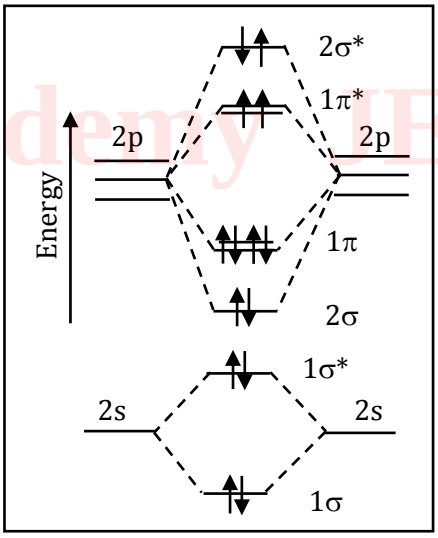

(B)




(C)




(D)




**CCB432**

**23.** Among $H_2$, $He_2^{\oplus}$, $Li_2$, $Be_2$, $B_2$, $C_2$, $N_2$, $O_2^{\ominus}$, and $F_2$, the number of diamagnetic species is :
(Atomic number : H = 1, He = 2, Li = 3, Be = 4, B = 5, C = 6, N = 7, O = 8 , F = 9)

[JEE (Advanced) 2017]

**CCB085**

**24.** According to Molecular Orbital Theory : [JEE (Advanced) 2016]

(A) $C_2^{2\ominus}$ is expected to be diamagnetic

(B) $O_2^{2\oplus}$ is expected to have a longer bond length than $O_2$

(C) $N_2^{\oplus}$ and $N_2^{\ominus}$ have the same bond order

(D) $He_2^{\oplus}$ has the same energy as two isolated He atoms

**CCB084**



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# ANSWER KEY

## EXERCISE (O-1)

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | A | B | A | B | B | C | B | C | D | D |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | D | A | C | A | B | A | D | A | D | D |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| Ans. | A | B | A | B | C | A | A | C | C | B |
| Que. | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| Ans. | B | C | C | A | B | C | A | B | B | B |
| Que. | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| Ans. | C | A | C | A | C | A | A | C | C | B |
| Que. | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| Ans. | A | A | C | C | C | D | B | B | B | C |
| Que. | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| Ans. | C | A | D | C | B | B | B | C | A | B |
| Que. | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| Ans. | C | B | C | C | B | B | A | D | C | A |
| Que. | 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| Ans. | C | C | B | A | C | C | B | C | A | D |
| Que. | 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |
| Ans. | A | A | D | C | B | D | C | B | A | A |
| Que. | 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 | 109 | 110 |
| Ans. | B | D | C | B | B | C | A | B | D | C |
| Que. | 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 | 119 | 120 |
| Ans. | A | C | C | B | C | D | D | C | B | B |
| Que. | 121 | 122 | 123 | 124 | 125 | 126 | 127 | 128 | 129 | 130 |
| Ans. | B | C | B | B | A | D | D | A | A | A |
| Que. | 131 | 132 | 133 | 134 | 135 | 136 | 137 | 138 | 139 | 140 |
| Ans. | C | B | A | B | B | D | B | B | C | B |
| Que. | 141 | 142 | 143 | 144 | 145 | 146 | 147 | 148 | 149 | 150 |
| Ans. | C | A | A | C | B | C | A | D | A | A |
| Que. | 151 | 152 | 153 | 154 | 155 | 156 | 157 | 158 | 159 | 160 |
| Ans. | A | D | D | A | B | C | A | B | B | C |
| Que. | 161 | 162 | 163 | 164 | 165 | 166 | 167 | 168 | 169 | 170 |
| Ans. | B | C | D | C | C | C | A | B | A | A |
| Que. | 171 | 172 | 173 | 174 | 175 | 176 | 177 | 178 | 179 | 180 |
| Ans. | A | C | A | D | D | A | B | B | C | A |
| Que. | 181 | 182 | 183 | 184 | 185 | 186 | 187 | 188 | 189 | 190 |
| Ans. | C | D | D | D | B | B | A | B | A | A |
| Que. | 191 | 192 | 193 | 194 | 195 | 196 | 197 | 198 | 199 | 200 |
| Ans. | C | B | C | B | B | B | D | D | D | A |

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## EXERCISE (O-2)

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | AC | AC | AC | BC | ABC | AB | CD | BC | BCD | BC |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | ABD | ABC | BCD | CD | ACD | BD | BCD | BCD | ABC | ABC |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| Ans. | AB | ABC | CD | ABCD | ACD | ABD | BD | ABC | ABC | ABC |
| Que. | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| Ans. | ABC | ACD | BD | ABCD | AB | AB | ABD | ABC | ABC | AD |
| Que. | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| Ans. | C | D | A | D | A | A | D | B | A | A |
| Que. | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| Ans. | D | A | C | B | C | B | C | C | B | C |

## EXERCISE (S)

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **1.** | 5 | **2.** | 4 | **3.** | 2 | **4.** | 5 | **5.** | 0 |
| **6.** | 1 | **7.** | 60 | **8.** | 3 | **9.** | 3 | **10.** | 3 |
| **11.** | 4 | **12.** | 4 | **13.** | 4 | **14.** | 7 | **15.** | 4 |
| **16.** | 6 | **17.** | 4 | **18.** | 16 | **19.** | 1 | **20.** | 4 |
| **21.** | 3 | **22.** | 2 | **23.** | 4 | **24.** | 6 | **25.** | 4 |
| **26.** | 3.00 | **27.** | 10.00 | **28.** | 4.58 | **29.** | 1.68-1.70 | | |
| **30.** | 95.00 | **31.** | 4.00 | **32.** | 0.62 | **33.** | 6.75 | | |
| **34.** | 4.00 | **35.** | 0.00 | | | | | | |

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## EXERCISE - JEE (Main) PYQ

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | 2 | 3 | 1 | 4 | 4 | 2 | 6 | 1 | 2 | 1 |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | 3 | 2 | 5 | 1 | 3 | 1 | 2 | 1 | 1 | 4 |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| Ans. | 1 | 2 | 1 | 4 | 3 | 1 | 8 | 1 | 2 | 1 |
| Que. | 31 | 32 | 33 | 34 | 35 | 36 | | | | |
| Ans. | 3 | 1 | 1 | 1 | 2 | 3 | | | | |

## EXERCISE - JEE (Advanced) PYQ

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | B | AD | 8 | 6 | B | B | 6 | BC | 4 | D |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | 0 | 6 | BD | ACD | BD | 6 | 4 | BC | AD | C |
| Que. | 21 | 22 | 23 | 24 | | | | | | |
| Ans. | C | C | 5 or 6 | AC | | | | | | |



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